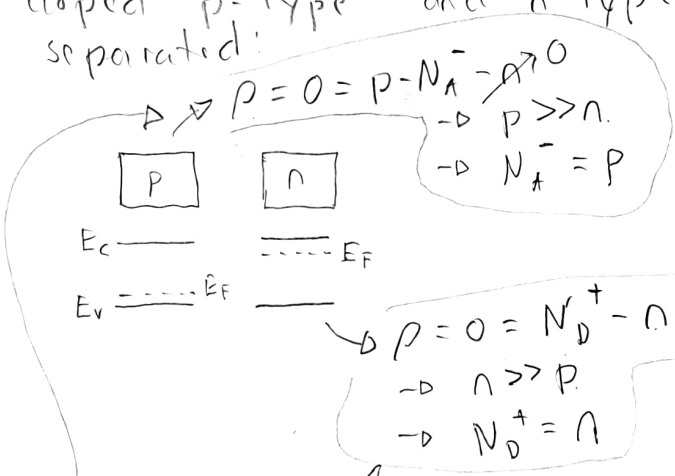
CHATER 3
p-n Junction

3.1 THERMAL EQUILIBRIUM CONDITION

1.1.1 Band Diagram

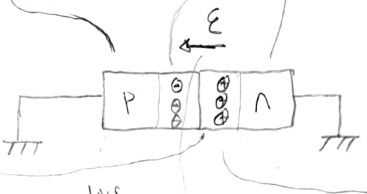
Uniformly doped p-type and n-type semiconductor physically separated:



$\rho = 0 = p - N_A^- - n \approx 0$
- $p \gg n$.
- $N_A^- = p$

$\rho = 0 = N_D^+ - n + p \approx 0$
- $n \gg p$
- $N_D^+ = n$

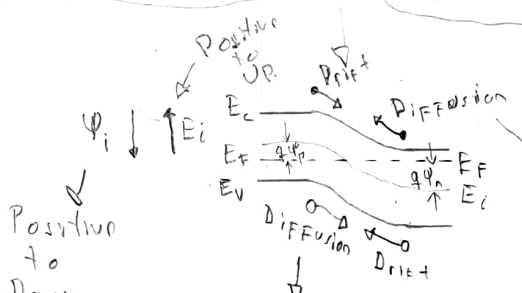
p-n junction at thermal equilibrium:



Metallurgical junction

$\rho = -N_A^- + p - n \approx 0$ (constant) — n increasing
- $p \approx n$

$\rho = +N_D^+ - p + n \approx 0$ — p increasing (q constant)
- $p \approx n$



From n-type to p-type

From p-type to n-type

At thermal equilibrium → No external excitations

Einstein's relation: $D_p = \frac{kT\mu_p}{q}$

Net hole current density:

$J_p = 0 = J_{p_{Drift}} + J_{p_{Diffusion}}$

$0 = (q\mu_p p \mathcal{E}) + (-qD_p \frac{dp}{dx})$

$0 = q\mu_p p\left(+\frac{1}{q}\frac{dE_i}{dx}\right) - q\left(\frac{kT\mu_p}{q}\right)\frac{dp}{dx}$

steady-state condition

$\mathcal{E} = -\nabla\psi_i$

$\frac{d\psi_i}{dx} = -\frac{1}{q}\frac{dE_i}{dx}$

$\mathcal{E} < 0$

$\frac{dp}{dx} < 0$

More p on p-side

$$0 = \mu_p p \frac{dE_i}{dx} - kT\mu_p \frac{dp}{dx} = 0$$

Hole concentration:

$p = n_i e^{(E_i - E_F)/kT}$ ← On every p-side.

$$\frac{dp}{dx} = p\left[\frac{d}{dx}\left(\frac{E_i}{kT}\right) - \frac{d}{dx}\left(\frac{E_F}{kT}\right)\right]$$

$$\frac{dp}{dx} = \frac{p}{kT}\left(\frac{dE_i}{dx} - \frac{dE_F}{dx}\right)$$

$$\frac{dE_i}{dx} = \frac{kT}{p}\frac{dp}{dx} + \frac{dE_F}{dx}$$

$$\rightarrow 0 = J_p = \mu_p p \frac{dE_i}{dx} - kT\mu_p \frac{dp}{dx} = 0$$

$$0 = \mu_p p\left(\frac{kT}{p}\frac{dp}{dx} + \frac{dE_F}{dx}\right) - kT\mu_p \frac{dp}{dx} = 0$$

$$\frac{dE_F}{dx} = 0$$

↑ On every p-side

Net electron current density:

$J_n = 0 = J_{n\,drift} + J_{n\,diffusion}$

$\rightarrow \mathcal{E} < 0$

$\rightarrow \frac{dn}{dx} > 0$

↑ More n on n-side

$$0 = +q\mu_n n\mathcal{E} + qD_n\frac{dn}{dx}$$

$$0 = q\mu_n n\left(\frac{1}{q}\frac{dE_i}{dx}\right) + q\left(\frac{kT}{q}\mu_n\right)\frac{dn}{dx}$$

Einstein's relation ← TD

$$D_n = \frac{kT}{q}\mu_n$$

Electron concentration:

$n = n_i e^{(E_F - E_i)/kT}$ ← Every n-side.

$$\frac{dn}{dx} = \frac{n}{kT}\left(\frac{dE_F}{dx} - \frac{dE_i}{dx}\right)$$

$$\rightarrow 0 = \mu_n n\left(-\frac{kT}{n}\frac{dn}{dx} + \frac{dE_F}{dx}\right) + kT\mu_n\frac{dn}{dx}$$

$$\frac{dE_i}{dx} = -\frac{kT}{n}\frac{dn}{dx} + \frac{dE_F}{dx}$$

$$\frac{dE_F}{dx} = 0$$ ← Every n-side

Thus on every charge space region: $\frac{dE_F}{dx} = 0$

$E_F \neq F(x)$

For the condition of zero net electron and hole currents ($J_p = 0 = J_n$)

Poisson's equation:

$$\nabla \vec{\mathcal{E}} = \frac{\rho_s}{\varepsilon_s}$$

$$-\frac{d\vec{\mathcal{E}}}{dx} = -\frac{\rho_s}{\varepsilon_s}$$

$$-\frac{d}{dx}\left(-\frac{d\psi}{dx}\right) = -\frac{\rho_s}{\varepsilon_s}$$

We assume that all donors and acceptors are ionized

$$\boxed{\frac{d^2\psi}{dx^2} = -\frac{\rho_s}{\varepsilon_s} = -\frac{q}{\varepsilon_s}\left(N_D^+ - N_A^- + p - n\right)}$$

In neutral regions:

$$\frac{d^2\psi}{dx^2} = 0 \quad \leftarrow \rho_s = 0$$

The electrostatic potential of the p-type neutral region with respect to the Fermi level: ψ_p:

$$q\psi_p = E_F - E_i \quad \leftarrow \psi_p < 0$$

$$q\psi_p = -kT\,Ln\left(\frac{p}{n_i}\right)$$

$$\psi_p = -\frac{kT}{q}\,Ln\left(\frac{N_A^-}{n_i}\right)$$

$$\boxed{p = n_i \exp\left(\frac{E_i - E_F}{kT}\right)}$$

$$\boxed{E_i - E_F = kT\,Ln\left(\frac{p}{n_i}\right)}$$

$\rightarrow N_D^+ = 0 \quad \rightarrow p \gg n$

$\rightarrow p = N_A^-$

The electrostatic potential of the n-type neutral region with respect to the Fermi level: ψ_n:

$$q\psi_n = E_F - E_i \rightarrow \psi_n > 0$$

$$\psi_n = \frac{kT}{q} Ln\left(\frac{n}{n_i}\right)$$

$$\psi_n = \frac{kT}{q} Ln\left(\frac{N_D^+}{n_i}\right)$$

$$n = n_i \exp\left(\frac{E_F - E_i}{kT}\right)$$

$$E_F - E_i = kT\, Ln\left(\frac{n}{n_i}\right)$$

$\rightarrow N_A^- = 0 \rightarrow n >> p$

$\rightarrow n = N_D^+$

Built-in potential V_{bi}:

$\psi_n > 0$ ← $\psi_p < 0$

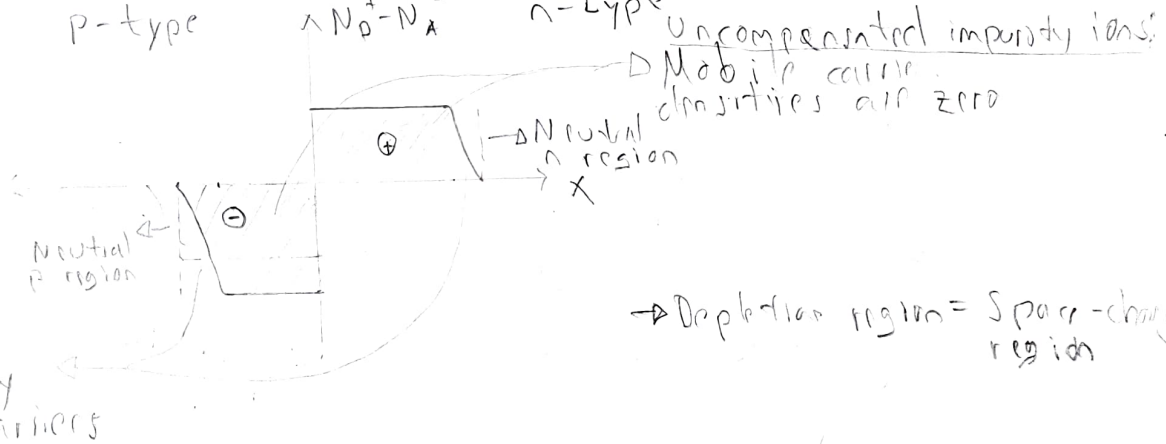
$$V_{bi} = |\psi_n| + |\psi_p| = \psi_n - \psi_p$$

$$V_{bi} = \psi_n - \psi_p = \frac{kT}{q}\left[Ln\left(\frac{N_D^+}{n_i}\right) + Ln\left(\frac{N_A^-}{n_i}\right)\right]$$

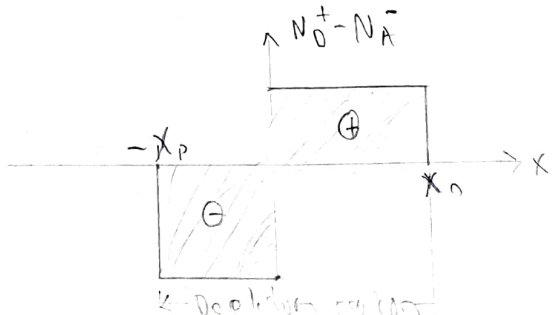
$$\boxed{V_{bi} = \frac{kT}{q} Ln\left(\frac{N_D^+ N_A^-}{n_i^2}\right) = \psi_n - \psi_p}$$

3.1.3 Space Charge

Space charge distribution: with abrupt doping changes



Transition region << Depletion Region → Rectangular of the space charge distribution



Poisson's equation on depletion region:

→ $p \approx n$

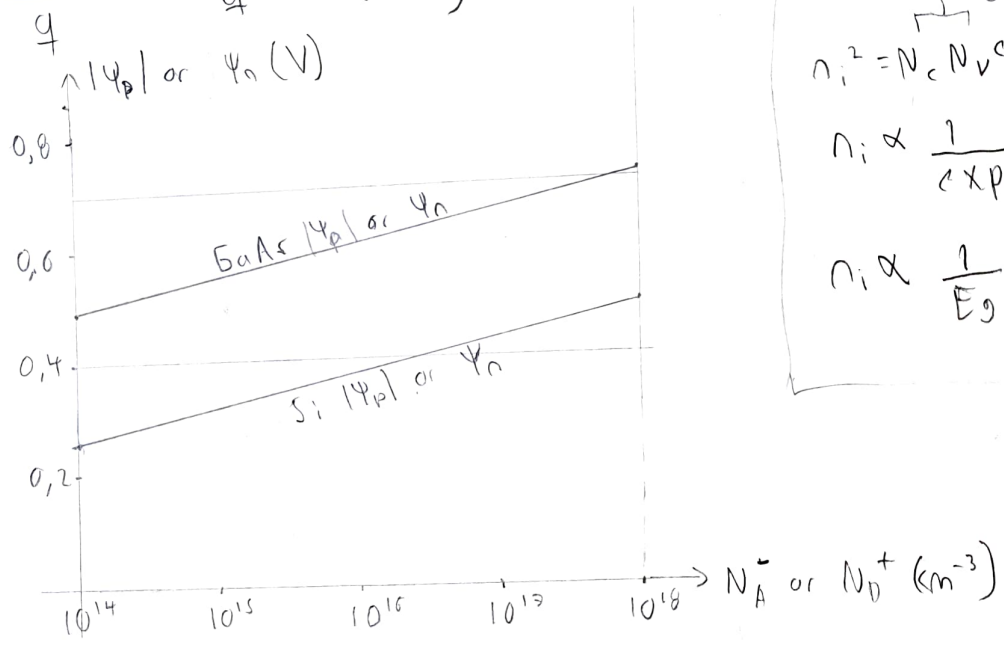
→ $\rho = N_D^+ - N_A^-$

→ $\frac{d^2\psi}{dx^2} = -\frac{\rho}{\varepsilon_s} = -\frac{q(N_D^+ - N_A^-)}{\varepsilon_s}$

$$\frac{d^2\psi}{dx^2} = \frac{q(N_A^- - N_D^+)}{\varepsilon_s}$$

From: $\psi_p = \frac{E_i - E_F}{q} = -\frac{kT}{q} \ln\left(\frac{N_A^-}{n_i}\right)$ } From neutral region

$\psi_n = \frac{E_F - E_i}{q} = \frac{kT}{q} \ln\left(\frac{N_D^+}{n_i}\right)$



→ $E_{g_{GaAs}} \approx 1,42$ eV → $E_{g_{Si}} \approx 1,12$ eV

$E_{g_{GaAs}} > E_{g_{Si}}$

↓

$n_{i_{GaAs}} < n_{i_{Si}}$

↓

$|\psi_p|$ or ψ_n > $|\psi_p|$ or ψ_n

GaAs Si

$E_{g_{GaAs}} > E_{g_{Si}}$

↓

$|\psi_p|$ or ψ_n > $|\psi_p|$ or ψ_n

GaAs Si

$n_i^2 = N_c N_v \exp\left(-\frac{E_g}{kT}\right)$ → Effective density of states

$n_i \propto \frac{1}{\exp\left(\frac{E_g}{kT}\right)}$

$n_i \propto \frac{1}{E_g}$

▷ EXAMPLE 1

$V_{bi} = ?$ $N_A = 10^{16}\,cm^{-3}$ $N_D = 10^{15}\,cm^{-3}$ at 300 K Silicon:

→ $n_i(300k) = 9,65^{+}10^{9}\,cm^{-3}$

For Silicon: Intrinsic carrier concentration

$$\rightarrow V_{bi} = \frac{kT}{q} Ln\left(\frac{N_A^- N_D^+}{n_i^2}\right)$$

$$V_{bi} = 0,0259V\, Ln\left(\frac{10^{16}\,10^{15}}{(9,65^{+}10^{9})^2}\right)$$

$$V_{bi} = 0,774V$$

From graph of Si: $|\Psi_p|$ or Ψ_n, V_s, N_A^- or N_D^+:

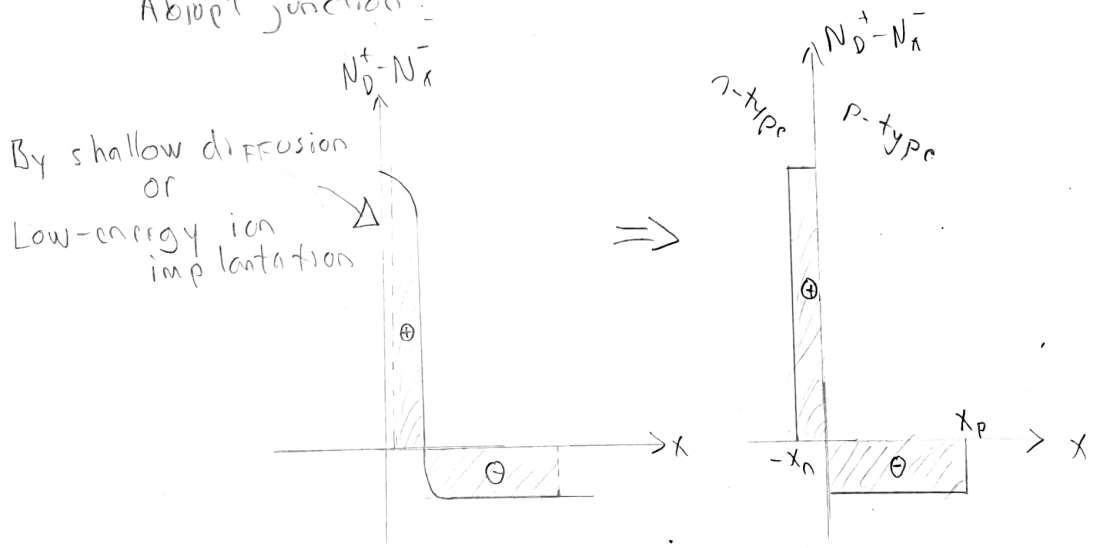
$$V_{bi} = \Psi_n - \Psi_p = \Psi_n + |\Psi_p| = \Psi_n(10^{15}cm^{-3}) + |\Psi_p|(10^{16}cm^{-3})$$

$$V_{bi} = (0,3V) + (0,47V) = 0,77V \checkmark$$

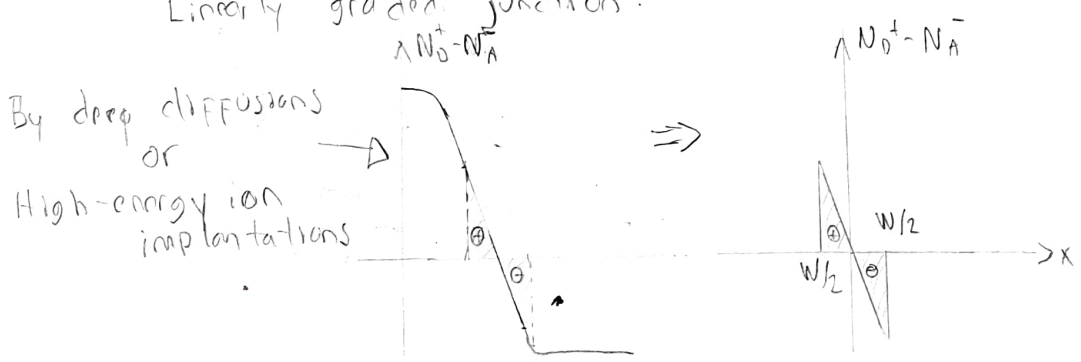
3.2 DEPLETION REGION

* Approximate doping profiles:

Abrupt junction:



Linearly graded junction:



3.2.1 Abrupt Junction

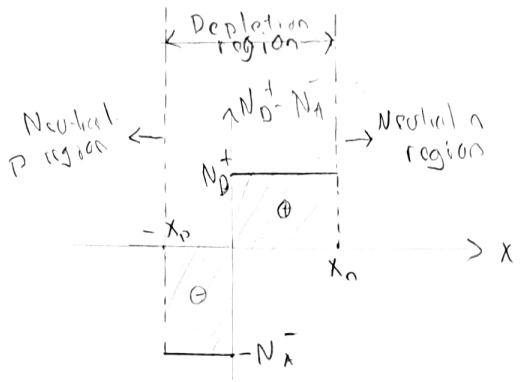
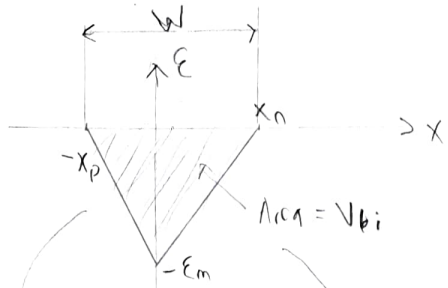
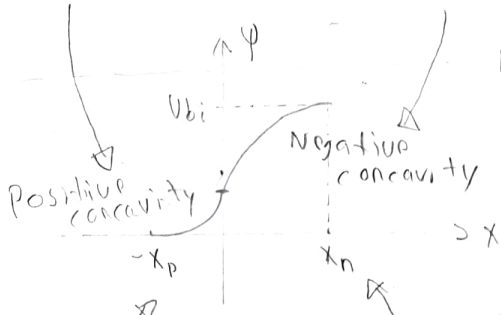


Fig. 6b:



slope < 0 slope > 0



$$\mathcal{E} = -\frac{d\psi}{dx}$$

$$\psi(x) = -\int_{-x_p}^{x} \mathcal{E}(x)\,dx \qquad \psi(x) = -\int_{x}^{x_n} \mathcal{E}(x)\,dx$$

$$\rightarrow \frac{d\psi}{dx} = -\mathcal{E}$$

$$\frac{d^2\psi}{dx^2} = -\frac{d\mathcal{E}}{dx}$$

Poisson equation:

For $-x_p \leq x < 0$:

$$\frac{d^2\psi}{dx^2} = -\frac{\rho}{\varepsilon_s} = -\frac{q(\cancel{N_D^+}^{\,0} - N_A^-)}{\varepsilon_s} = \frac{qN_A^-}{\varepsilon_s} = -\frac{d\mathcal{E}}{dx}$$

For $0 < x \leq x_n$:

$$\frac{d^2\psi}{dx^2} = -\frac{d\mathcal{E}}{dx} = -\frac{\rho}{\varepsilon_s} = -\frac{q(N_D^+ - \cancel{N_A^-}^{\,0})}{\varepsilon_s}$$

$$\frac{d^2\psi}{dx^2} = -\frac{qN_D^+}{\varepsilon_s} = -\frac{d\mathcal{E}}{dx}$$

The overall space charge neutrality

$$\cancel{A_{neg}} + N_A x_p = N_D x_n + \cancel{A_{neg}}$$

Depp.

$$N_A x_p = N_D x_n$$

$$\downarrow$$

$$N_A \propto \frac{1}{x_p} \quad \text{and} \quad N_D \propto \propto \frac{1}{x_n}$$

Total depletion layer width:

$$W = x_p + x_n$$

The electric field:

$$\frac{d\psi}{dx} = -\mathcal{E}$$

$$\mathcal{E} = -\frac{d\psi}{dx}$$

$$\mathcal{E} = -\int \frac{d^2\psi}{dx^2} dx$$

From Fig. 6.b

For $-x_p \leq x < 0$:

$$\int_0^{\mathcal{E}} d\mathcal{E} = -\int_{-x_p}^{x} \frac{q N_A^-}{\varepsilon_s} dx$$

$$\mathcal{E} = -\frac{q N_A^-}{\varepsilon_s}(x + x_p)$$

$$\mathcal{E} = -\frac{q N_A^-}{\varepsilon_s} x - \frac{q N_A^- x_p}{\varepsilon_s}$$ Positive

$$\mathcal{E} = -\frac{q N_A^- x}{\varepsilon_s} - \mathcal{E}_m$$ When $x = 0$

From Fig 6.b

For $0 < x \leq x_n$

$$\int_{\mathcal{E}}^{0} d\mathcal{E} = -\int_{x}^{x_n} \frac{q N_D^+}{\varepsilon_s} dx$$

$$\mathcal{E} = \int_{x_n}^{x} \frac{q N_D^+}{\varepsilon_s} dx$$

$$\mathcal{E} = \frac{q N_D^+}{\varepsilon_s}(x - x_n)$$

$$\mathcal{E} = \frac{q N_D^+}{\varepsilon_s} x - \frac{q N_D^+}{\varepsilon_s} x_n$$

$$\mathcal{E} = \frac{q N_D^+}{\varepsilon_s} x - \mathcal{E}_m$$ When $x = 0$

$\mathcal{E}_m$ is the maximum field that exists at $x=0$:

$$\mathcal{E}_m = \mathcal{E}(x=0)$$

$$\mathcal{E}_m = \frac{q N_A^- x_p}{\varepsilon_s} = \frac{q N_D^+ x_n}{\varepsilon_s}$$

$$N_A^- x_p = N_D^+ x_n \checkmark$$

Total potential variation: Built-in potential V_{bi}:

At: $-x_p \le x < 0$:

$$d\psi = -\mathcal{E}\,dx$$

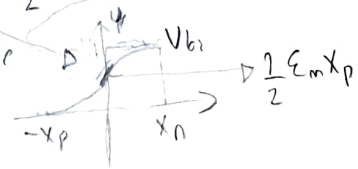
$$\int_0^{\psi} d\psi = -\int_{-x_p}^{x} \left(-\frac{q N_A^- x}{\varepsilon_s} - \mathcal{E}_m\right) dx$$

$$\psi = \frac{q N_A^-}{2\varepsilon_s} x^2 \Big|_{-x_p}^{x} + \mathcal{E}_m x \Big|_{-x_p}^{x}$$

$$\psi = \frac{q N_A^-}{2\varepsilon_s}\left(x^2 - x_p^2\right) + \mathcal{E}_m (x + x_p)$$

$$\psi = \frac{q N_A^-}{2\varepsilon_s} x^2 - \frac{\mathcal{E}_m x_p}{2} + \mathcal{E}_m x + \mathcal{E}_m x_p$$

$$\int_0^{\psi} d\psi = \underbrace{\frac{q N_A^-}{2\varepsilon_s} x^2}_{\text{Positive concavity}} + \mathcal{E}_m x + \frac{1}{2}\mathcal{E}_m x_p$$



Thus:

$$\psi(-x_p) = \frac{q N_A^-}{2\varepsilon_s} x_p^2 - \mathcal{E}_m x_p + \frac{1}{2}\mathcal{E}_m x_p$$

$$\psi(-x_p) = \frac{1}{2}\mathcal{E}_m x_p - \frac{1}{2}\mathcal{E}_m x_p \checkmark$$

At: $0 < x \le x_n$

$$\int_{V_{bi}}^{\psi} d\psi = -\int_x^{x_n} \left(\frac{q N_D^+}{\varepsilon_s} x - \mathcal{E}_m\right) dx$$

$$\int_{V_{bi}}^{\psi} d\psi = \int_x^{x_n} \left(\frac{q N_D^+}{\varepsilon_s} x - \mathcal{E}_m\right) dx$$

$$\psi = V_{bi} + \frac{q N_D^+ x_n^2}{2\varepsilon_s} - \frac{q N_D^+ x^2}{2\varepsilon_s} - \mathcal{E}_m (x_n - x)$$

$$\psi = V_{bi} - \frac{q N_D^+ x^2}{2\varepsilon_s} + \frac{\mathcal{E}_m}{2} x_n - \mathcal{E}_m x_n + \mathcal{E}_m x$$

$$\psi = \underbrace{-\frac{q N_D^+}{2\varepsilon_s} x^2}_{\text{Negative concavity}} + \mathcal{E}_m x + \left(V_{bi} - \frac{\mathcal{E}_m}{2} x_n\right)$$

See box next page

$$\int_{V_{bi}}^{\psi} d\psi = -\frac{q N_D^+}{2\varepsilon_s} x^2 + \mathcal{E}_m x - \frac{\mathcal{E}_m}{2} x_n$$

Thus:

$$\psi(x_n) = -\frac{q N_D^+}{2\varepsilon_s} x_n^2 + \mathcal{E}_m x_n + V_{bi} - \frac{\mathcal{E}_m}{2} x_n$$

$$\psi(x_n) = -\frac{\mathcal{E}_m}{2} x_n + \frac{1}{2}\mathcal{E}_m x_n + V_{bi}$$

$$\psi(x_n) = V_{bi} \checkmark$$

$$V_{bi} = -\int_{x_p}^{x_n} \mathcal{E}\,dx = \left(-\int_{-x_p}^{0} \mathcal{E}\,dx\right) + \left(-\int_{0}^{x_n} \mathcal{E}\,dx\right)$$

$$V_{bi} = \int_{0}^{\psi(x=0)} d\psi + \int_{\psi(x=0)}^{V_{bi}} d\psi$$

$$V_{bi} = -\int_{-x_p}^{x=0} \mathcal{E}\,dx - \int_{V_{bi}}^{\psi(x=0)} d\psi$$

$$V_{bi} = \frac{qN_A^- \cancel{x^2}^{0}}{2\varepsilon_s} - \varepsilon_m \cancel{x}^{0} + \frac{\varepsilon_m}{2} x_p - \left(-\frac{qN_D^+ \cancel{x^2}^{0}}{2\varepsilon_s} + \varepsilon_m \cancel{x}^{0} - \frac{\varepsilon_m}{2} x_n\right)$$

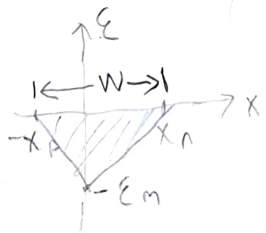
$$V_{bi} = \frac{\varepsilon_m}{2} x_p + \frac{\varepsilon_m x_n}{2} \longrightarrow$$

At $0 < x \leq x_n$:

$$\psi = -\frac{qN_D^+ x^2}{2\varepsilon_s} + \varepsilon_m x + \left(V_{bi} - \frac{\varepsilon_m}{2} x_n\right)$$

$$V_{bi} = \frac{\varepsilon_m}{2}(x_p + x_n)$$

$$V_{bi} = \frac{\varepsilon_m}{2} W \xrightarrow{A=1}$$



$$\psi = -\frac{qN_D^+ x^2}{2\varepsilon_s} + \varepsilon_m x + \frac{\varepsilon_m x_p}{2}$$

Total depletion layer with $W(V_{bi}, N_A^-, N_D^+)$:

$$\rightarrow V_{bi} = \frac{qN_A^- x_p^2}{2\varepsilon_s} + \frac{qN_D^+ x_n^2}{2\varepsilon_s} \rightarrow x_n = \frac{N_A^- x_p}{N_D^+}$$

$$x_n = \sqrt{\frac{2\varepsilon_s V_{bi}}{q(N_D^+ + N_A^-)}\left(\frac{N_A}{N_D}\right)}$$

$$\rightarrow V_{bi} = \frac{qN_D x_n^2}{2\varepsilon_s} + \frac{qN_A x_p^2}{2\varepsilon_s} \rightarrow x_p = \frac{N_D x_n}{N_A}$$

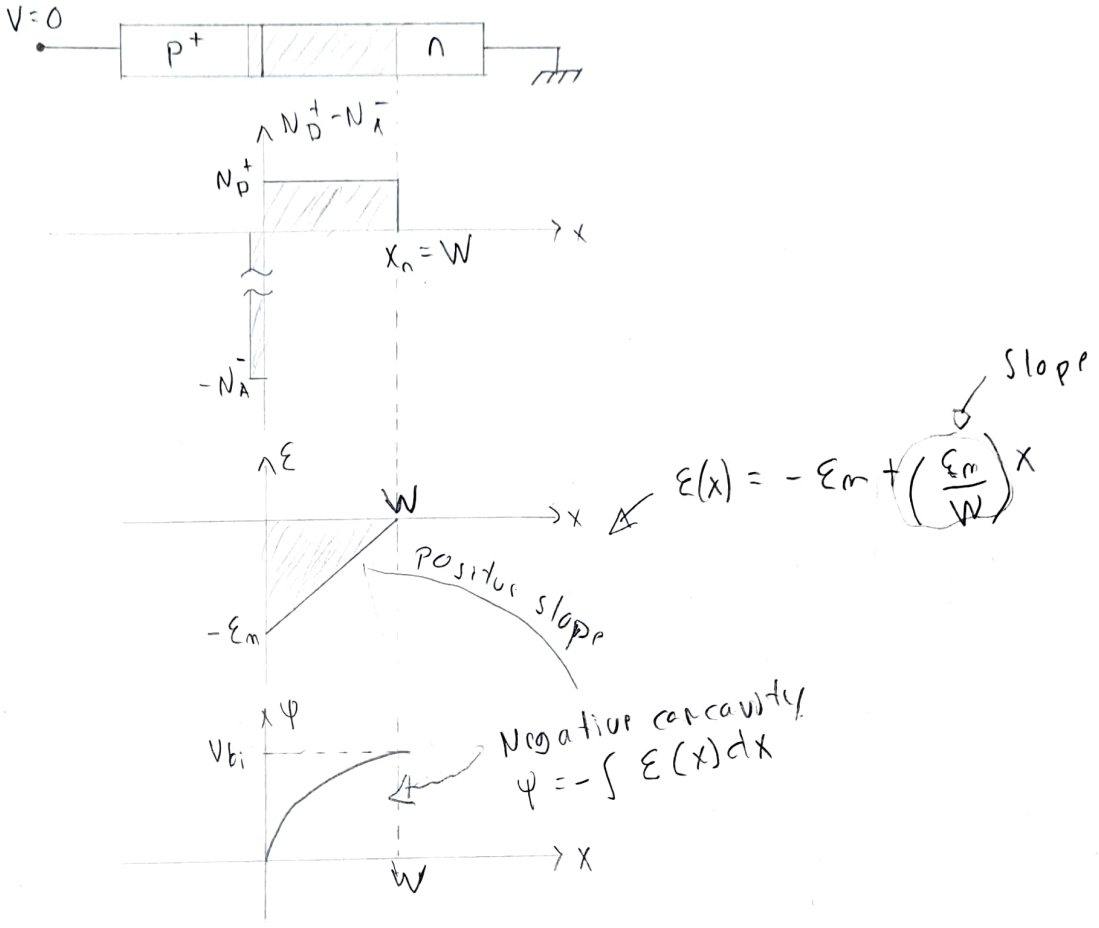
$$x_p = \sqrt{\frac{2\varepsilon_s V_{bi}}{q(N_D + N_A)}\left(\frac{N_D}{N_A}\right)}$$

$$\rightarrow W = x_n + x_p$$

$$W = \sqrt{\frac{2\varepsilon_s V_{bi}}{q(N_D + N_A)}}\left(\frac{\sqrt{N_A}}{\sqrt{N_D}} + \frac{\sqrt{N_D}}{\sqrt{N_A}}\right)$$

$$W = \sqrt{\frac{2\varepsilon_s}{q}\left(\frac{N_A + N_D}{N_A N_D}\right) V_{bi}}$$ ← Only abrupt junction

* One-sided abrupt junction ($N_A^- \gg N_D^+$) in thermal equilibrium:



Slope

$$\mathcal{E}(x) = -\mathcal{E}_m + \left(\frac{\mathcal{E}_m}{W}\right)x$$

Space charge neutrality:

$$N_A x_p = N_D x_n$$

$$x_p = \frac{N_D}{N_A} x_n \rightarrow N_A \gg N_D \rightarrow x_p \ll x_n$$

Depletion layer width of abrupt junction:

$$W = \sqrt{\frac{2\varepsilon_s}{q}\left(\frac{N_A + N_D}{N_A N_D}\right) V_{bi}}$$

$$W \approx x_n = \sqrt{\frac{2\varepsilon_s}{q N_D} V_{bi}}$$

Electric-field distribution: ($0 < x \leq x_n$):

$$\mathcal{E}(x) = -\mathcal{E}_m + \frac{q N_D}{\varepsilon_s} x$$

Lightly doped bulk concentration: N_B

p^+-n junction → $N_B = N_D$

$$\mathcal{E}(x) = -\mathcal{E}_m + \frac{qN_B}{\varepsilon_s}x$$

→ $$\mathcal{E}(x_n) = -\mathcal{E}_m + \frac{qN_B}{\varepsilon_s}x_n$$

$$0 = -\mathcal{E}_m + \frac{qN_D}{\varepsilon_s}x_n$$

$$\mathcal{E}_m = \frac{qN_B}{\varepsilon_s}x_n$$

$$\boxed{\mathcal{E}_m = \frac{qN_D}{\varepsilon_s}W}$$

→ $$\mathcal{E}(x) = -\mathcal{E}_m + \frac{qN_B}{\varepsilon_s}x$$

$$\mathcal{E}(x) = -\frac{qN_BW}{\varepsilon_s} + \frac{qN_B}{\varepsilon_s}x$$

$$\mathcal{E}(x) = -\frac{qN_B}{\varepsilon_s}(W - x)\frac{W}{W}$$

$$\mathcal{E}(x) = -\frac{qN_BW}{\varepsilon_s}\left(+1 - \frac{x}{W}\right)$$

$$\mathcal{E}(x) = -\mathcal{E}_m\left(1 - \frac{x}{W}\right) = -\mathcal{E}_m + \frac{\mathcal{E}_m}{W}x$$

slope

→ Poisson's equation:

$$\int_{\psi=0}^{\varphi} \psi\, d\psi = -\int_0^x \mathcal{E}\, dx$$

With zero potential in the neutral p-region as a reference



$$\varphi = -\int_0^x \left(-\mathcal{E}_m + \frac{\mathcal{E}_m}{W}x\right)dx$$

$$\psi = \mathcal{E}_m x - \frac{\mathcal{E}_m x^2}{2W} = \mathcal{E}_m x\left(1 - \frac{x}{2W}\right) = \frac{\mathcal{E}_m x}{2}\left(2 - \frac{x}{W}\right)$$

From: $V_{bi} = \frac{\mathcal{E}_m W}{2}$ ← Every abrupt junction

$$\frac{\mathcal{E}_m}{2} = \frac{V_{bi}}{W}$$

Thus:

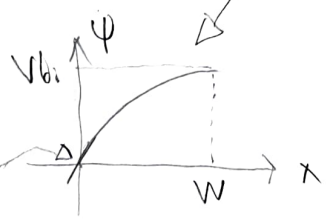
$$\psi = \left(\frac{V_{bi}}{W}\right) x \left(2 - \frac{x}{W}\right)$$

$$\boxed{\psi = \frac{V_{bi}\, x}{W}\left(2 - \frac{x}{W}\right)}$$

$$\psi(x) = \frac{2V_{bi}}{W} x - \frac{V_{bi} x^2}{W^2}$$

slope

Potential distribution



$$\psi(W) = V_{bi}$$

EXAMPLE 2

One-sided abrupt junction: $N_A^- = 10^{19}\,cm^{-3}$, $N_D^+ = 10^{16}\,cm^{-3} \rightarrow N_B$

$W = ?$ → at $T = 300K$ → $n_i = 9.65 * 10^9\,cm^{-3}$

$\mathcal{E}_m = ?$

$$\rightarrow V_{bi} = \frac{kT}{q} Ln\left(\frac{N_A N_D}{n_i^2}\right) = (0.0259V)\, Ln\left[\frac{(10^{19}cm^{-3})(10^{16}cm^{-3})}{(9.65 * 10^9 cm^{-3})^2}\right]$$

$$V_{bi} = 0.896V$$

$$\rightarrow W \approx x_n = \sqrt{\frac{2\varepsilon_s}{q N_B} V_{bi}} = \sqrt{\frac{2(11.9 * 553 * 10^3\, \cancel{e}/V cm)}{\cancel{e}\,(10^{16}\,cm^{-3})}(0.896V)}$$

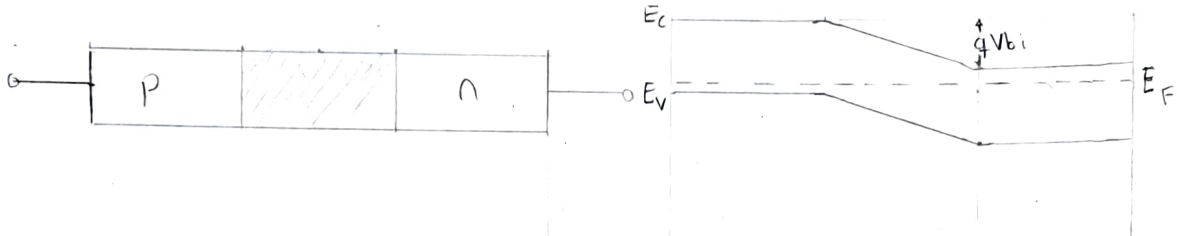
$$W = 34.3 * 10^{-6}\,cm = 34.3 * 10^{-6}(10^9\,nm) = 343\,nm$$

$$\rightarrow \mathcal{E}_m = \frac{q N_B W}{\varepsilon_s} = \frac{\cancel{e}\,(10^{16}\,cm^{-3})(34.3 * 10^{-6}\,cm)}{(11.9 * 553 * 10^3\, \cancel{e}/V cm)} =$$

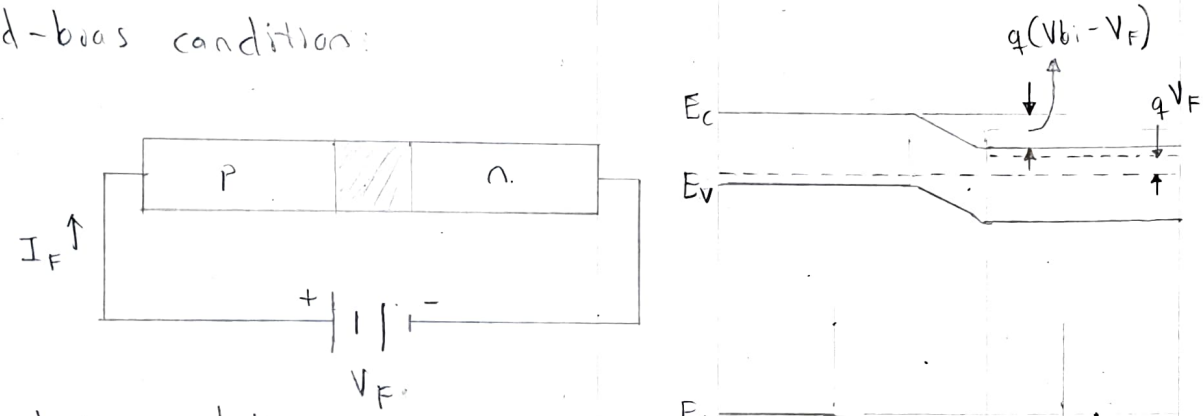
$$\mathcal{E}_m = 52.7 * 10^3\, V/cm$$

*p-n junction under various biasing conditions:

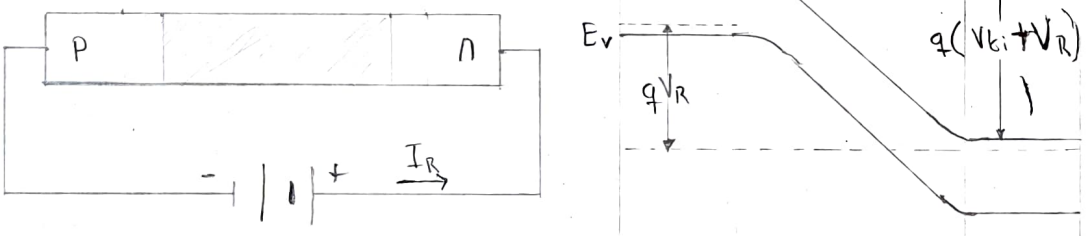
a) Thermal-equilibrium condition:



b) Forward-bias condition:



c) Reverse-bias condition



Depletion layer widths as a function of the applied voltage for a one-side abrupt junction:

$$W = \sqrt{\frac{2\varepsilon_s(V_{bi}-V)}{qN_B}}$$

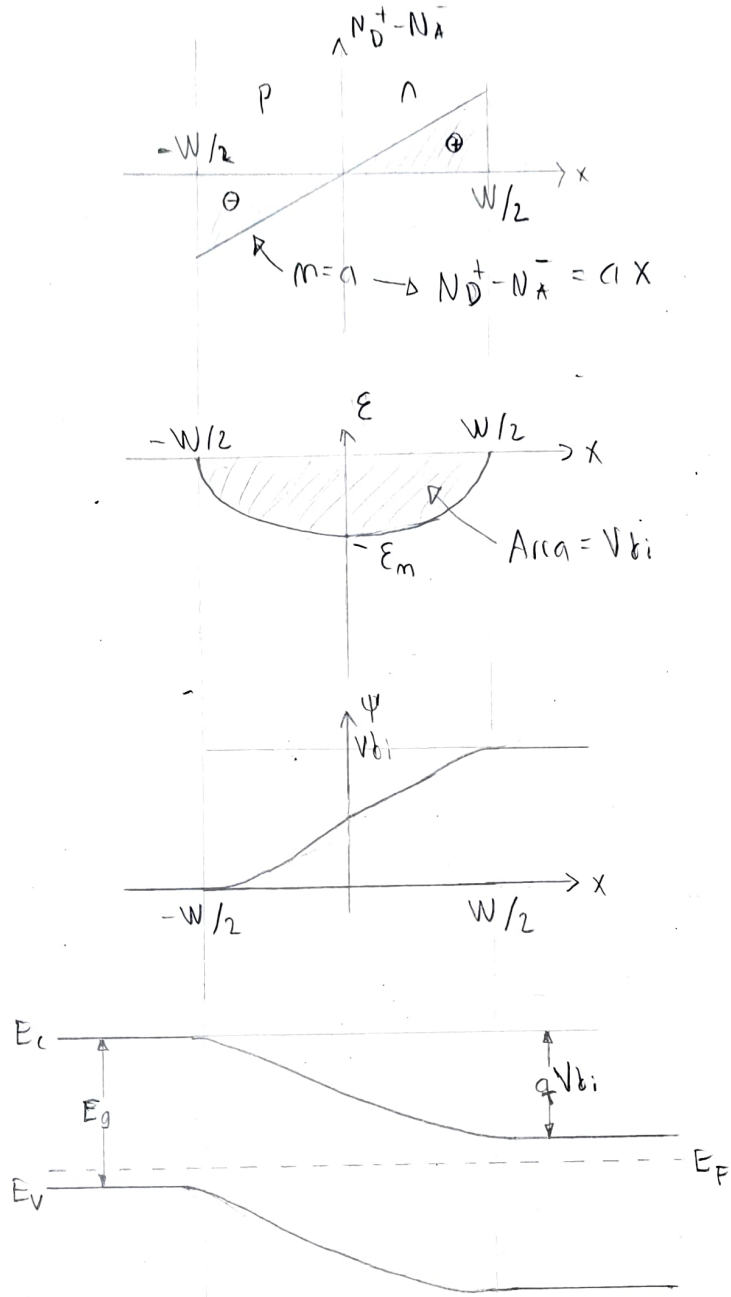
$V > 0 \leftarrow$ Forward bias

$V < 0 \leftarrow$ Reverse bias

$$\downarrow$$

$$W \propto \sqrt{V_{bi}-V}$$

3.2.2. Linearly Graded Junction



The Poisson's equation:

$$\frac{d^2\psi}{dx^2} = -\frac{d\mathcal{E}}{dx} = \frac{-\rho_s}{\varepsilon_s} = \frac{-q}{\varepsilon_s} ax \qquad -\frac{W}{2} \leq x \leq \frac{W}{2}$$

a = Impurity gradient

$[a] = cm^{-4}$

Mobile carriers are negligible in the depletion region:

$$p \approx 0$$

$$n \approx 0$$

Electric Field distribution:

$$\frac{d\mathcal{E}}{dx} = \frac{q\,a\,x}{\varepsilon_s}$$

$$\int_0^{\mathcal{E}} d\mathcal{E} = \int_{-W/2}^{x} \frac{q\,a\,x}{\varepsilon_s}\,dx \qquad -\frac{W}{2} \le x \le 0$$

$$\mathcal{E} = \frac{q\,a}{2\varepsilon_s}\left[x^2 - (-W/2)^2\right]$$

$$\mathcal{E} = -\frac{q\,a}{2\varepsilon_s}\left[(W/2)^2 - x^2\right]$$

For $0 \le x \le W/2$

$$\int_{\mathcal{E}}^{0} d\mathcal{E} = \int_{x}^{W/2} \frac{q\,a\,x}{\varepsilon_s}\,dx \qquad \text{Equal}$$

$$-\mathcal{E} = \frac{q\,a}{2\varepsilon_s}\left[(W/2)^2 - x^2\right]$$

$$\mathcal{E} = -\frac{q\,a}{2\varepsilon_s}\left[(W/2)^2 - x^2\right]$$

$$\mathcal{E} = \frac{q\,a}{2\varepsilon_s}x^2 - \frac{q\,a}{2\varepsilon_s}\left(\frac{W}{2}\right)^2$$

$\rightarrow m(x=0) = 0$

At $x = 0$

$$\mathcal{E}_m = \frac{q\,a\,W^2}{8\varepsilon_s}$$

Potential distribution:

$$\mathcal{E} = -\frac{d\psi}{dx}$$

$$d\psi = -\mathcal{E}\,dx$$

$$\int_0^{\psi} d\psi = -\int_{-W/2}^{x} \left(-\frac{qa}{2\varepsilon_s}\left[(W/2)^2 - x^2\right]\right) dx$$

$$\psi(x) = \frac{qa}{2\varepsilon_s}\left(-\frac{x^3}{3} + \frac{W^2}{4}x + \frac{W^3}{12}\right)$$

At $x = -W/2$

$$\psi(-W/2) = \frac{qa}{2\varepsilon_s}\left[-\left(-\frac{W}{2}\right)^3\frac{1}{3} + \frac{W^2}{4}\left(-\frac{W}{2}\right) + \frac{W^3}{12}\right]$$

$$\psi(-W/2) = 0 \quad \checkmark$$

At $x = W/2$

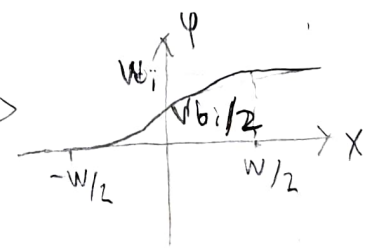
$$\psi(W/2) = V_{bi} = \frac{qa}{2\varepsilon_s}\left[-\left(\frac{W}{2}\right)^3\frac{1}{3} + \frac{W^2}{4}\left(\frac{W}{2}\right) + \frac{W^3}{12}\right]$$

$$V_{bi} = \frac{qaW^3}{12\varepsilon_s}$$

At $x = 0$

$$\psi(0) = \frac{qa}{2\varepsilon_s}\left(\frac{W^3}{12}\right)$$

$$\psi(x=0) = \frac{V_{bi}}{2}$$



Depletion layer width:

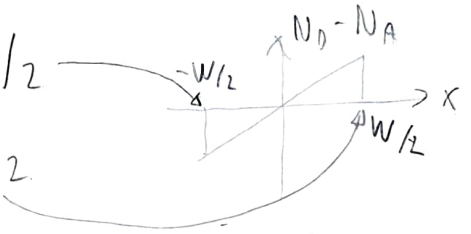
$$V_{bi} = \frac{q a W^3}{12 \varepsilon_s}$$

$$W = \left(\frac{12 \varepsilon_s V_{bi}}{q a} \right)^{1/3}$$

Values of impurity concentrations at the edges of the depletion region: ($x = -W/2$ and $x = W/2$)

$$N_A^- = aW/2$$

$$N_D^+ = aW/2$$



From:

$$V_{bi} = \frac{KT}{q} Ln\left(\frac{N_A^- N_D^+}{n_i^2} \right) = \frac{KT}{q} Ln\left[\left(\frac{aW}{2n_i} \right)^2 \right]$$

$$V_{bi} \approx 2\frac{KT}{q} Ln\left[\frac{aW}{2n_i} \right]$$

Based on an accurate numerical technique:

$$V_{bi} = \frac{2}{3}\frac{KT}{q} Ln\left(\frac{a^2 \varepsilon_s KT/q}{8q n_i^3} \right)$$

But:

$$0.05V < \frac{2KT}{q} Ln\left(\frac{aW}{2n_i} \right) - \frac{2KT}{3q} Ln\left(\frac{a^2 \varepsilon_s KT/q}{8q n_i^3} \right) < 0.1V$$

$$\Downarrow$$

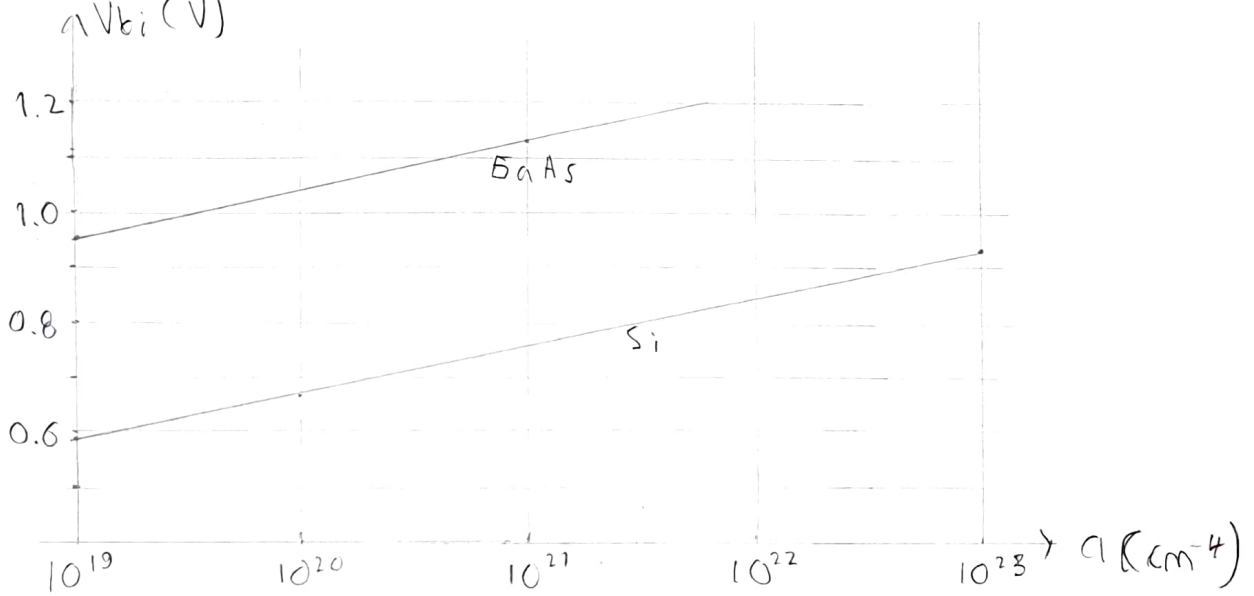
$$V_{bi} \approx \frac{2KT}{q} Ln\left(\frac{aW}{2n_i} \right)$$ is a good aproximation

$$V_{bi} = \frac{2kT}{q} \ln\left(\frac{aW}{2n_i}\right)$$

$$V_{bi} = \frac{2kT}{q} \ln\left[\frac{a}{2n_i}\left(\frac{12\varepsilon_s V_{bi}}{qa}\right)^{1/3}\right]$$

Solving this transcendental equation for:

$V_{bi}(a) \neq F(W)$:



When either forward or reverse bias, depletion layer width:

$$W = \left(\frac{12\varepsilon_s(V_{bi}-V)}{qa}\right)^{1/3}$$

$W \propto (V_{bi}-V)^{1/3}$ $\quad V>0 \leftarrow$ Forward bias

$V<0 \leftarrow$ Reverse bias

EXAMPLE 3

Si (T=300K) $\quad a = 10^{20}\,cm^{-4}$ $\quad W=?,\ \varepsilon_m=?,\ V_{bi}=?$

If $V_{bi} = 0.3V$: ← Trial 1

$$W = \left(\frac{12\varepsilon_s V_{bi}}{qa}\right)^{1/3} = \left[\frac{12(11.9 * 553 * 10^3\ \not{F}/\not{cm})(0.3\ \not{V})}{\not{e}(10^{20}\ cm^{-\not{4}-3})}\right]^{1/3}$$

$W = 6.19 * 10^{-5}\ cm$

And:

$$V_{bi} = \frac{2kT}{q} \ln\left(\frac{aW}{2n_i}\right)$$

$$V_{bi} = 2(0.0259V) \ln\left[\frac{(10^{20}cm^{-4})(61.9*10^{-5}cm)}{2(9.65*10^{9}cm^{-3})}\right]$$

$V_{bi} = 0.657V \neq V_{bi} = 0.3V$ ← Trial 1

By means of $V_{bi} = 0.659V$ in:

$$W = \left(\frac{12\varepsilon_s V_{bi}}{qa}\right)^{1/3} \rightarrow V_{bi} = \frac{2kT}{q} \ln\left(\frac{aW}{2n_i}\right)$$ } Trial 2

trial	1	2	3	4
$V_{bi}(V)$	0.3	0.657	0.671	0.671
$W(cm)$	$6.19*10^{-5}$	$8.03*10^{-5}$	$8.09*10^{-5}$	$8.09*10^{-5}$
$V_{bi}(V)$	0.657	0.67	0.671	0.671

Thus: $W = 8.09*10^{-5}(10^4 \mu m) = 0.809 \mu m$

$V_{bi} = 0.671V$

$$\rightarrow \varepsilon_m = \frac{qaW^2}{8\varepsilon_s} = \frac{e(10^{20}cm^{-4})(8.09*10^{-5}cm)^2}{8(11.9*553*10^{3}e/Vcm)}$$

$\varepsilon_m = 1.24*10^4 V/cm$

Beside, from graph $V_{bi}(a) \rightarrow V_{bi}(10^{20}cm^{-3}) \approx 0.68V$ ✓ (Silicon)

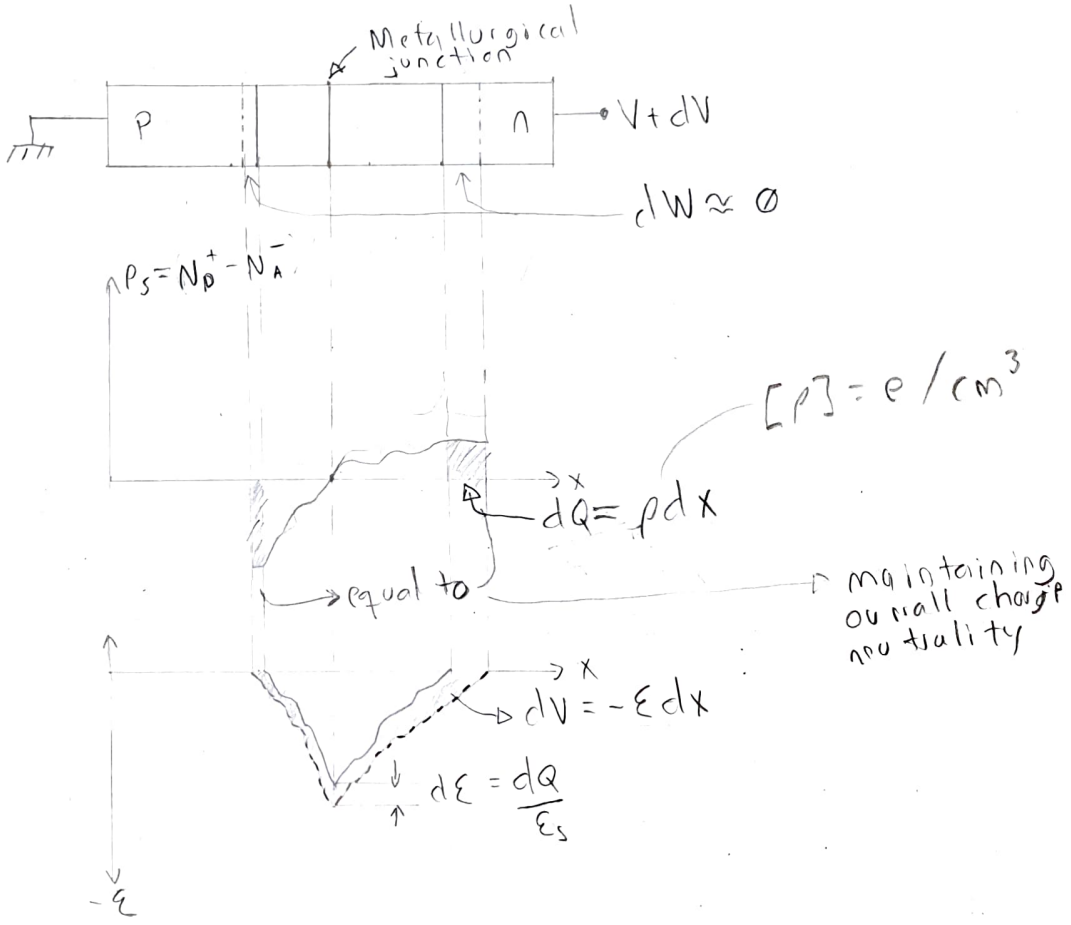
3.3 DEPLETION CAPACITANCE.

Junction depletion-layer capacitance per unit area:

$C_j = \frac{dQ}{dV}$ — dQ: the incremental charge in depletion-layer charge per unit area.
dV: incremental change in the applied voltage.

→ Also referred to as the transition region capacitance

p-n junction with an arbitrary impurity profile under reverse bias:



Incremental charge dQ causes an increase of $\mathcal{E}$:

$$\nabla \mathcal{E} = \frac{\rho}{\varepsilon_s}$$

$$\frac{d\mathcal{E}}{dx} = \frac{\rho}{\varepsilon_s}$$

$$d\mathcal{E} = \frac{\rho dx}{\varepsilon_s}$$

$$d\mathcal{E} = \frac{dQ}{\varepsilon_s}$$

Too:

$$dV = W d\mathcal{E}$$

$$dV = W \frac{dQ}{\varepsilon_s}$$

$$C_j = \frac{dQ}{dV} = \frac{dQ}{W \frac{dQ}{\varepsilon_s}} = \frac{\varepsilon_s}{W}$$

→ $F = [\varepsilon_s]\frac{cm^2}{cm}$ → $[C_j] = \frac{[\varepsilon_s]}{[W]}$ → $F = \frac{Coulomb}{V}$

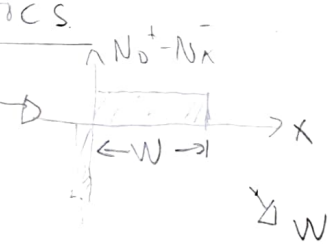
$$[\varepsilon_s] = \frac{F}{cm}$$

$$[C_j] = \frac{F}{cm^2} \checkmark = \frac{[e]}{V\,cm^2}$$ (e → electron)

$$\rightarrow \boxed{C_j = \frac{\varepsilon_s}{W}}$$

3.3.1 Capacitance-Voltage characteristics

For a one-sided abrupt junction:



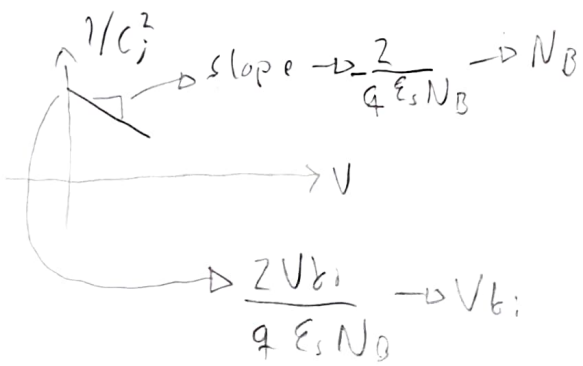
$$C_j = \frac{\varepsilon_s}{W} = \sqrt{\frac{\varepsilon_s^2}{\frac{2\varepsilon_s(V_{bi}-V)}{q N_B}}}$$

$$W = x_n = \sqrt{\frac{2\varepsilon_s(V_{bi}-V)}{q N_B}}$$

$$C_j = \sqrt{\frac{\varepsilon_s q N_B}{2(V_{bi}-V)}}$$

$$\boxed{\frac{1}{C_j^2} = \frac{2(V_{bi}-V)}{q\varepsilon_s N_B}}$$

$$\frac{1}{C_j^2} = \frac{2V_{bi}}{q\varepsilon_s N_B} - \left(\frac{2}{q\varepsilon_s N_B}\right)V$$



EXAMPLE 4

$N_A = 2*10^{19}\,cm^{-3}$
$N_D = 8*10^{15}\,cm^{-3}$
} $N_B = 8*10^{15}\,cm^{-3}$

$C_j(V=0) = ?$ $T = 300°K$

$C_j(V=-4V) = ?$

↳ Reverse bias

→ $V_{bi} = \frac{KT}{q} Ln\left(\frac{N_D N_A}{n_i^2}\right)$ → For every homojunction

$$V_{bi} = 0.0259V \cdot Ln\left[\frac{(8*10^{15}cm^{-3})(2*10^{19}cm^{-3})}{(9.65*10^{9}(cm^{-3})^2}\right]$$

$V_{bi} = 0.909V$

Abrupt junction:

→ $W(V=0) = \sqrt{\frac{2\varepsilon_s}{q}\left(\frac{N_A + N_D}{N_A N_D}\right)V_{bi}}$

$$W = \sqrt{\frac{2\varepsilon_s V_{bi}}{q N_B}}$$

$$W = \sqrt{\frac{2(11.9*553E3\ e/V cm)(0.909V)}{e\,(8E15 cm^{-3})}}$$

$W = 3.87E-5\ cm = 3.87E-5\,(1E4\,\mu)\,m$

$W = 0.387\,mm$

→ $C_j(V=0) = \frac{\varepsilon_s}{W(V=0)} = \frac{\sqrt{\varepsilon_s^2}}{\sqrt{\frac{2\varepsilon_s V_{bi}}{q N_B}}} = \sqrt{\frac{q\varepsilon_s N_B}{2V_{bi}}}$

$$C_j(V=0) = \frac{(11.9*553*10^{3}\ e/V cm)}{3.87*10^{-5} cm}$$

→ $F = \frac{[q]}{V} = \frac{Coulomb}{V}$

$C_j = 1.7*10^{11}\frac{e}{V\,cm^2}$

$C_j = 1.7*10^{11}\frac{(1.6*10^{-19}\,C)}{V\,cm^2}$

$C_j = 27.2\frac{nF}{cm^2}$

For reverse bias: $V = -4V$

$$W(V=-4V) = \sqrt{\frac{2\varepsilon_s(V_{bi}-V)}{qN_B}}$$

$$W(V=-4V) = \sqrt{\frac{2(11.9 * 553 * 10^3\ eV/Vcm)[0.909V + 4V]}{e(8*10^{15}\ cm^{-3})}}$$

$$W = 8.99 * 10^{-5}\ cm = 8.99 * 10^{-1}\ \mu m = 0.899\ \mu m$$

$$\rightarrow C_j(V=-4V) = \frac{\varepsilon_s}{W(V=-4)} = \sqrt{\frac{q\varepsilon_s N_B}{2(V_{bi}-V)}}$$

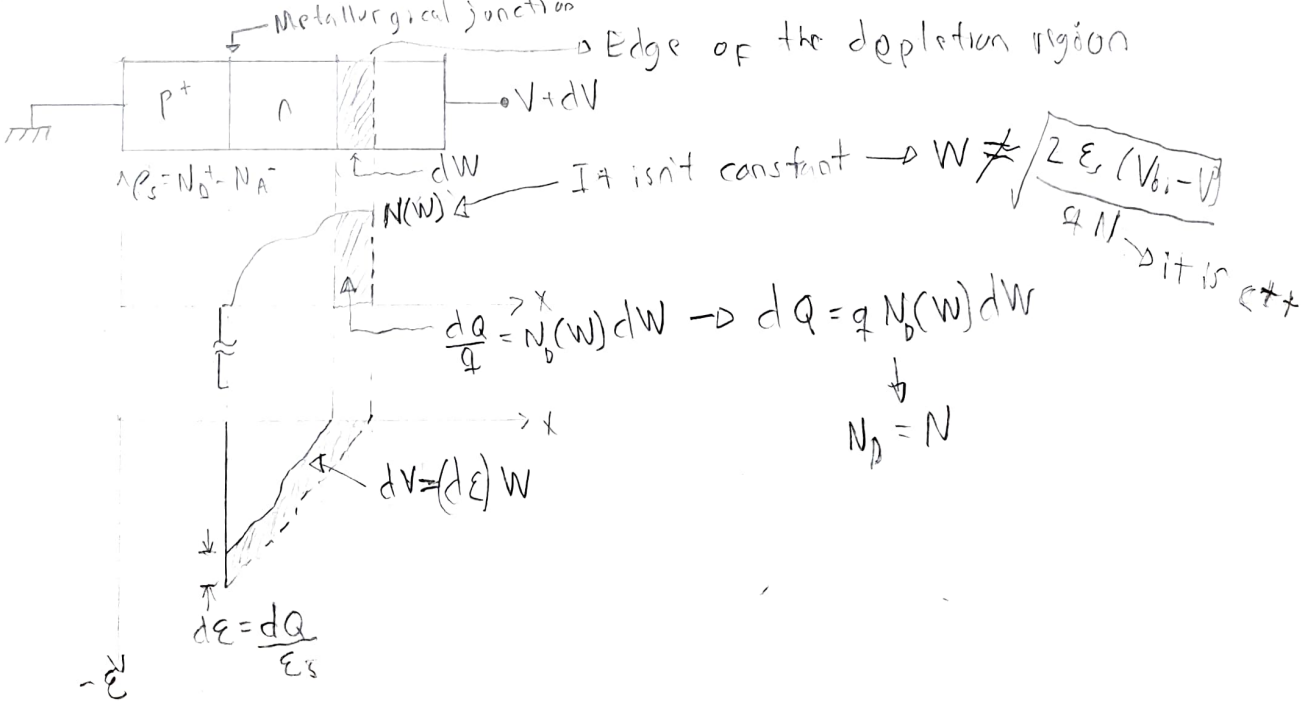
$$C_j = \frac{(11.9 * 553 * 10^3\ e/Vcm)}{8.99 * 10^{-5}\ cm}$$

$$C_j = 7.32 * 10^{10}\ \frac{e}{V\,cm^2} = 7.32 * 10^{10}\ \frac{(1.6 * 10^{-19}\ C)}{V\,cm^2}$$

$$C_j = 1.17 * 10^{-8}\ F/cm^2 = 11.7\ nF/cm^2$$

3.3.2 Evaluation of Impurity Distribution

*p^+-n junction with an arbitrary impurity distribution at reverse bias:



Change in applied voltage:

$$dV = (d\mathcal{E})W = \left(\frac{dQ}{\varepsilon_s}\right)W$$

$$dV = \frac{(qN(W)\,dW)\,W}{\varepsilon_s}$$

$$dV = \frac{qN(W)\,d(W^2)}{2\varepsilon_s}$$

$\rightarrow d\left(\frac{W^2}{2}\right) = W\,dW$

$$\rightarrow d(W^2) = d\left(\frac{\varepsilon_s^2}{C_j^2}\right) = \varepsilon_s^2\, d\left(\frac{1}{C_j^2}\right)$$

Impurity concentration at the edge of the depletion region:

$$N(W) = \frac{2}{q\varepsilon_s}\left[\frac{1}{d(1/C_j^2)/dV}\right]$$

slope of $\frac{1}{C_j^2}$ Vs V

impurity profile $\{N(W)$ ← $W = \frac{\varepsilon_s}{C_j}$

* For a linearly graded junction:

$$\rightarrow W = \left(\frac{12\varepsilon_s(V_{bi}-V)}{qa}\right)^{1/3}$$

a → Impurity gradient

$$\rightarrow C_j = \frac{\varepsilon_s}{W}$$

$$C_j = \left[\frac{qa\varepsilon_s^2}{12(V_{bi}-V)}\right]^{1/3}$$

Coulomb

$$[C_j] = \left(\frac{C\,cm^{-4}\,F^2/cm^2}{V}\right)^{1/3}$$

$$= \left(\frac{C\,cm^{-6}}{V}\left(\frac{C}{V}\right)^2\right)^{1/3}$$

$$= \left(\frac{C^3}{V^3}\right)^{1/3} cm^{-2} = F\,cm^{-2}$$ ✓

$$\frac{1}{C_j^3} = -\underbrace{\left(\frac{12}{q a \varepsilon_s^2}\right)}_{\text{slope}} V + \underbrace{\frac{12 V_{bi}}{q a \varepsilon_s^2}}_{\text{intercept}}$$

slope → a

intercept → V_{bi}

3.3.3 Varactor

Variable reactor → Varactor

For abrupt junction at reverse-biased depletion:

$$C_j = \sqrt{\frac{q \varepsilon_s N_B}{2(V_{bi} - V)}}$$

For a linearly graded junction at reverse-biased depletion:

$$C_j = \left[\frac{q a \varepsilon_s^2}{12(V_{bi} - V)}\right]^{1/3}$$

$$\rightarrow V_R = -V$$

Thus:

$C_j \propto (V_{bi} + V_R)^{-n}$ → $n = 1/3$ → linearly graded junction

$C_j \propto V_R^{-n}$ → $n = 1/2$ → Abrupt junction

$$\frac{\partial C_j}{\partial V_R} \propto -n V_R^{-1-n}$$

Variation of C with V_R is greater for an abrupt junction

→ $-1-\frac{1}{3} = -\frac{4}{3}$ → $-1-\frac{1}{2} = -\frac{3}{2}$

→ $-\frac{1}{3} > -\frac{1}{2}$ → $-\frac{4}{3} > -\frac{3}{2}$

$-8 > -9$

I DON'T KNOW

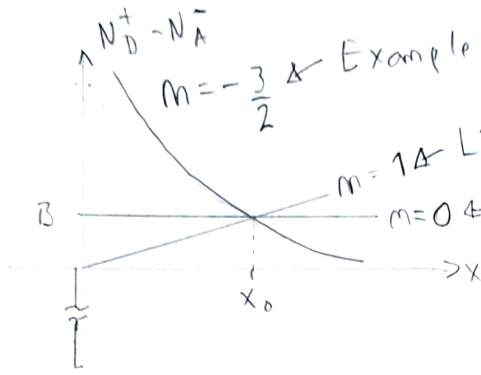
→ $\frac{\partial C_j}{\partial V_R} \propto -\frac{1}{3} V_R^{-4/3}$ ← linearly graded

→ $\frac{\partial C_j}{\partial V_R} \propto -\frac{1}{2} V_R^{-3/2}$ ← Abrupt

Hyperabrupt junction: $\Lambda > 1/2$

Impurity profiles for p^+-n doping:

$$N_D(x) = B\left(\frac{x}{x_0}\right)^m \rightarrow B \text{ and } x_0 \text{ are constants}$$



↖ Epitaxial growth techniques

To obtain the C; Vs V:

$$\rightarrow \frac{d^2\varphi}{dx^2} = \frac{-q}{\varepsilon_s}\left(N_D^+ - N_A^-\right)$$

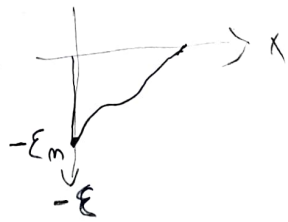
If B have q and ε_s, at $x > 0$:

$$\frac{d^2\psi}{dx^2} = -B\left(\frac{x}{x_0}\right)^m \qquad \rightarrow \mathcal{E} = -\frac{d\psi}{dx}$$

$$\frac{d}{dx}(-\mathcal{E}) = -B\left(\frac{x}{x_0}\right)^m$$

$$\frac{d\mathcal{E}}{dx} = B\left(\frac{x}{x_0}\right)^m$$

From p^+-n:



$$\mathcal{E} = \int B\left(\frac{x}{x_0}\right)^m dx + C_1$$

$$\mathcal{E} = \frac{B x^{m+1}}{(m+1) x_0^m} + C_1$$

$$\mathcal{E} = \frac{B x^{m+1}}{(m+1) x_0^m} - \mathcal{E}_m$$

At $x = W$

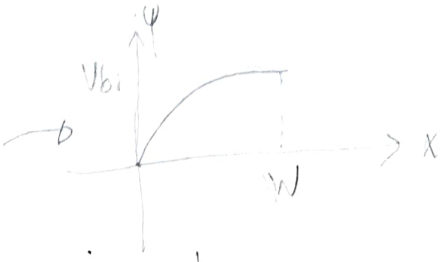
$$\rightarrow 0 = \frac{B W^{m+1}}{(m+1) x_0^m} - \mathcal{E}_m$$

$$\mathcal{E}_m = \frac{B W^{m+1}}{(m+1) x_0^m}$$

$$\rightarrow \mathcal{E} = -\frac{d\psi}{dx}$$

$$\rightarrow -\frac{d\psi}{dx} = \frac{B x^{m+1}}{(m+1) x_0^m} - \mathcal{E}_m$$

$$\psi = -\frac{B x^{m+2}}{(m+1)(m+2) x_0^m} + \mathcal{E}_m x + C_2$$



$$\rightarrow C_2 = 0$$

$$\psi = -\frac{B x^{m+2}}{(m+1)(m+2) x_0^m} + \mathcal{E}_m x$$

At thermal equilibrium:

$$V_{bi} = -\frac{B W^{m+2}}{(m+1)(m+2) x_0^m} + \mathcal{E}_m W$$

$$V_{bi} = -\frac{B W^{m+2}}{(m+1)(m+2) x_0^m} + \left[\frac{B W^{m+1}}{(m+1) x_0^m}\right] W$$

$$V_{bi} = \frac{B W^{m+2}}{(m+2) x_0^m}\left(1 - \frac{1}{m+2}\right)$$

$m \geq 0$

$$\rightarrow 1 - \frac{1}{m+2} > 0$$

Without thermal equilibrium:

$$V_{bi} + V_R = \frac{B W^{m+2}}{(m+1) x_0^m}\left(1 + \frac{1}{m+2}\right)$$

$$W \propto V_R^{1/(m+2)}$$

$$\rightarrow C_j = \frac{\varepsilon_s}{W} \propto V_R^{-1/(m+2)} \qquad \rightarrow C_j \propto V_R^{-n}$$

$$\rightarrow n = \frac{1}{m+2}$$

more hyperabrupt

$$m = \frac{1}{n} - 2 \rightarrow n \uparrow \rightarrow m \downarrow$$

more negative

For hyperabrupt junction with $m = -\frac{3}{2}$

$$n = \frac{1}{-\frac{3}{2} + 2}$$

$$n = \frac{2}{-3+4} = 2$$

$$n = 2 > \frac{1}{2}$$

When the varactor $n=2$ is connected to an inductor L, the resonant frequency:

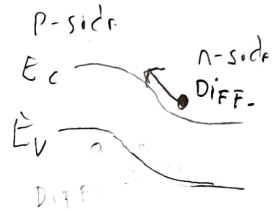
$$\omega_r = \frac{1}{\sqrt{LC_j}} \propto \frac{1}{\sqrt{V_R^{-n}}}$$

For $n = 2$:

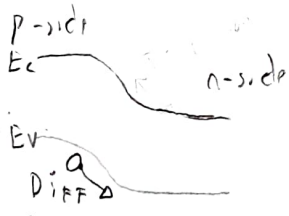
$$\omega_r \propto V_R$$

3.4 CURRENT-VOLTAGE CHARACTERISTICS

* Under forward bias:
  - Voltage → Reduce the electrostatic potential across the depletion region
  - Electrons of the n-side CB, have enough energy
    ↓
    surmount the barrier
    ↓
    Diffuse from the n-side to p-side



  - holes of the p-side VB, have enough energy
    ↓
    surmount the barrier
    ↓
    Diffuse from the p-side to n-side



* Under reverse bias:

→ Voltage → Increase the electrostatic potential across the depletion region.

↓

Reduce the diffusion currents

* For the drift current:

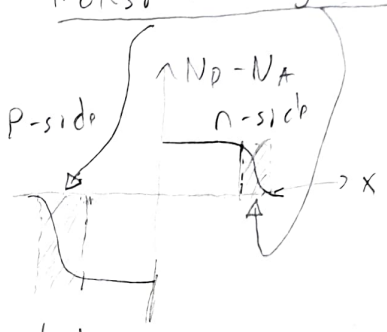
Minoritary electrons and holes into the transition regions:

↓

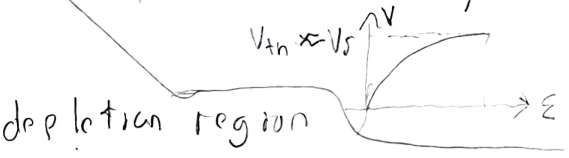
Drift current depends mainly on the number of minoritary carriers.

↓

Drift current no depend of barrier change.



↘ Travel at almost their saturation velocity



* Drift and diffusion current → In the depletion region

↓

Difficult to derive the current equations

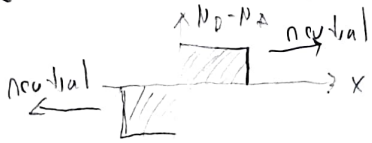
↓

Current equations by diffusions outside the depletion region

3.4.1 Ideal Characteristics.

* Ideal current-voltage characteristics:

→ Depletion regions has abrupt boundaries:



→ Low-injection condition → Injected minoritary carrier << Majoritary carrier

→ No generation-recombination currents in depletion region

→ No current within the depletion region

At thermal equilibrium in every neutral region:

$\rightarrow N_A = p_{p0}$ $\qquad$ $\rightarrow N_D = n_{n0}$

↑ Thermal equilibrium

Built-in potential:

$$V_{bi} = \frac{kT}{q} \ln\left(\frac{p_{p0} n_{n0}}{n_i^2}\right) = \frac{(E_{F_{n0}} - E_i) + (E_i - E_{F_{p0}})}{q}$$

Mass action law: $p_{p0} n_{p0} = n_i^2 = p_{n0} n_{n0}$

$$\rightarrow V_{bi} = \frac{kT}{q} \ln\left(\frac{p_{p0} n_{n0}}{p_{p0} n_{p0}}\right) \qquad \rightarrow V_{bi} = kT \ln\left(\frac{p_{p0} n_{n0}}{p_{n0} n_{n0}}\right)$$

$$n_{n0} = n_{p0} e^{qV_{bi}/kT} \qquad p_{p0} = p_{n0} e^{qV_{bi}/kT}$$

minority carriers

When a forward and reverse bias at the boundaries of the depletion region:

$$n_n = n_p e^{q(V_{bi}-V)/kT} \qquad p_p = p_n e^{q(V_{bi}-V)/kT}$$

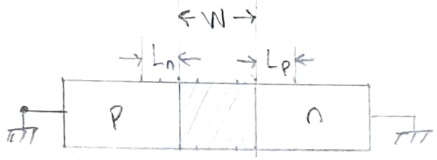
For the low-injection condition: $n_n \approx n_{n0}$, $p_p \approx p_{p0}$:

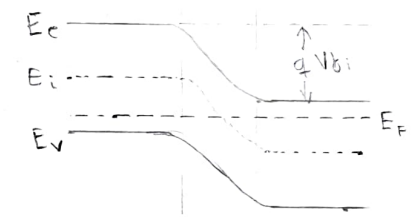
$$n_p = \frac{n_{n0}}{e^{q(V_{bi}-V)/kT}} \qquad p_n = \frac{p_{p0}}{e^{q(V_{bi}-V)/kT}}$$

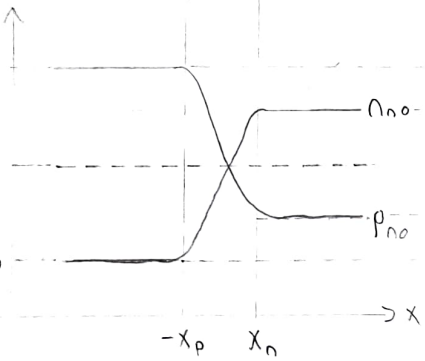
$$n_p = \frac{\left(n_{p0} e^{qV_{bi}/kT}\right)}{e^{q(V_{bi}-V)/kT}} \qquad p_n = \frac{\left(p_{n0} e^{qV_{bi}/kT}\right)}{e^{q(V_{bi}-V)/kT}}$$

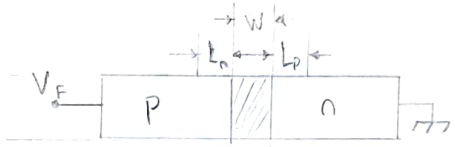
At $-x_p$: $n_p = n_{p0} e^{qV/kT}$ $\qquad$ $p_n = p_{n0} e^{qV/kT}$ ↙ At $+x_n$

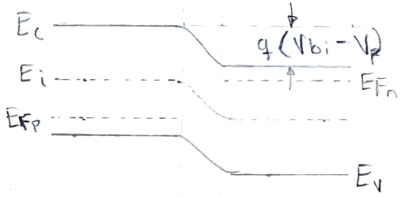
$\rightarrow n_p - n_{p0} = n_{p0}\left(e^{qV/kT} - 1\right)$ $\qquad$ $p_n - p_{n0} = p_{n0}\left(e^{qV/kT} - 1\right)$









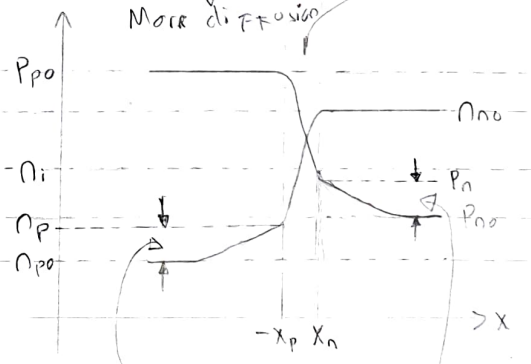


$\rightarrow V = V_F$

Narrower depletion width

Forst slope of carrier distribution

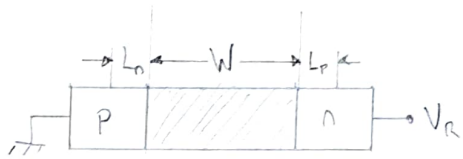
More diffusion

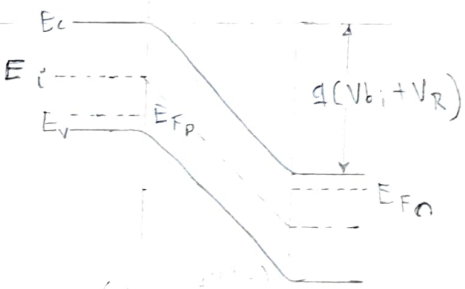


$n_{po}(e^{qV/KT} - 1)$

$P_{no}(e^{qV/KT} - 1)$

↳ Injected minority carriers recombine with mayoritary carriers

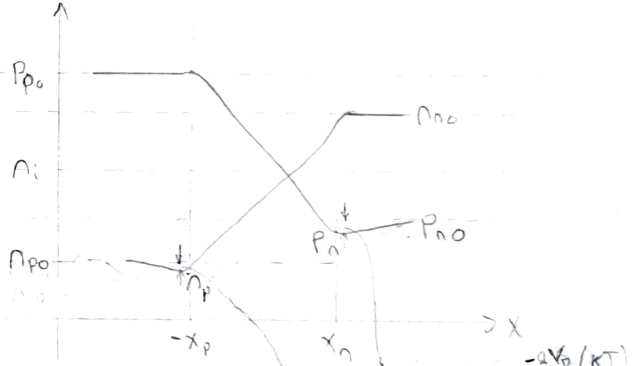




$\rightarrow n_{po} - n_p = n_{po}(1 - e^{qV/KT})$

$\rightarrow P_{no} - P_n = P_{no}(1 - e^{qV/KT})$

$\rightarrow V = -V_R$



$\rightarrow P_{no}(1 - e^{-qV_R/KT})$

$\rightarrow n_{po}(1 - e^{-qV_R/KT})$

Steady-state continuity equation at LOW INJECTION:

$$\frac{\partial p_n}{\partial t}^{\to 0} = -p_n \mu_p \frac{\partial \mathcal{E}}{\partial x} - \mathcal{E}\mu_p \frac{\partial p_n}{\partial x} + D_p \frac{\partial^2 p_n}{\partial x^2} + G_{Lp} - \frac{p_n - p_{no}}{\tau_p}$$

In the neutral n-region: $(x \gtrsim x_n)$

$$0 = -p_n \mu_p \frac{\partial \mathcal{E}}{\partial x}^{\to 0} - \mathcal{E}\mu_p \frac{\partial p_n}{\partial x}^{\to 0} + D_p \frac{\partial^2 p_n}{\partial x^2} + G_{Lp}^{\to 0} - \frac{p_n - p_{no}}{\tau_p}$$

$\to p_n(x \to \infty) = p_{no}$

$\to L_p = \sqrt{D_p \tau_p}$

$$\frac{d^2 p_n}{dx^2} - \frac{p_n}{L_p^2} = -\frac{p_{no}}{L_p^2} \qquad \to p_n(x \to \infty) = p_{no}$$

Homogeneous solution:

$$p_n = A e^{-x/L_p} + B e^{+x/L_p}$$

Complete solution:

$$p_n = A e^{-x/L_p} + B e^{x/L_p} + p_{no}$$

$$p_{no} = \lim_{x \to \infty} \left(A e^{-x/L_p} + B^{\to 0} e^{x/L_p} + p_{no} \right)$$

$$p_{no} = p_{no}$$

At $x = x_n$: $p_n = p_{no} e^{qV/kT}$

$$p_{no} e^{qV/kT} = A e^{-x_n/L_p} + p_{no}$$

$$A = \frac{p_{no}(e^{qV/kT} - 1)}{e^{-x_n/L_p}}$$

$$\to \quad p_n = \left[\frac{p_{no}(e^{qV/kT} - 1)}{e^{-x_n/L_p}} \right] e^{-x/L_p} + p_{no}$$

$$p_n - p_{no} = p_{no}(e^{qV/kT} - 1)\, e^{-(x - x_n)/L_p}$$

At $x = x_n$:

$$J_p(x = x_n) = -q D_p \underbrace{\left.\frac{dp_n}{dx}\right|_{x=x_n}}_{\text{negative in fig.}} = \frac{q D_p}{L_p} p_{no}(e^{qV/kT} - 1) e^{-(x - x_n)/L_p}\Big|_{x=x_n}^{1}$$

Steady-state continuity equation at LOW-INJECTION:

$$\frac{\partial n_p}{\partial t}^{\to 0} = n_p \mu_n \frac{\partial \mathcal{E}}{\partial x} + \mathcal{E} \mu_p \frac{\partial n_p}{\partial x} + D_n \frac{\partial^2 n_p}{\partial x^2} + G_{Ln} - \frac{n_p - n_{po}}{\tau_n}$$

In the neutral p-region: $(x \leq -x_p)$

$$D_n \frac{d^2 n_p}{dx^2} - \frac{n_p - n_{po}}{\tau_n} = 0 \qquad \to L_n = \sqrt{D_n \tau_n}$$

$$\frac{d^2 n_p}{dx^2} - \frac{n_p}{L_n^2} = -\frac{n_{po}}{L_n^2}$$

Complete solution:

$$n_p = A e^{-x/L_n} + B e^{+x/L_n} + n_{po}$$

Since $n_p(x \to -\infty) = n_{po}$: $A = 0$

$$n_p = B e^{x/L_n} + n_{po}$$

At $x = -x_p$: $n_p = n_{po} e^{qV/kT}$

$$n_{po} e^{qV/kT} = B e^{-x_p/L_n} + n_{po}$$

$$B = \frac{n_{po}\left(e^{qV/kT} - 1\right)}{e^{-x_p/L_n}}$$

So: $n_p = n_{po}\left(e^{qV/kT} - 1\right) e^{(x+x_p)/L_n} + n_{po}$

$$n_p - n_{po} = n_{po}\left(e^{qV/kT} - 1\right) e^{(x+x_p)/L_n}$$

At $x = -x_p$

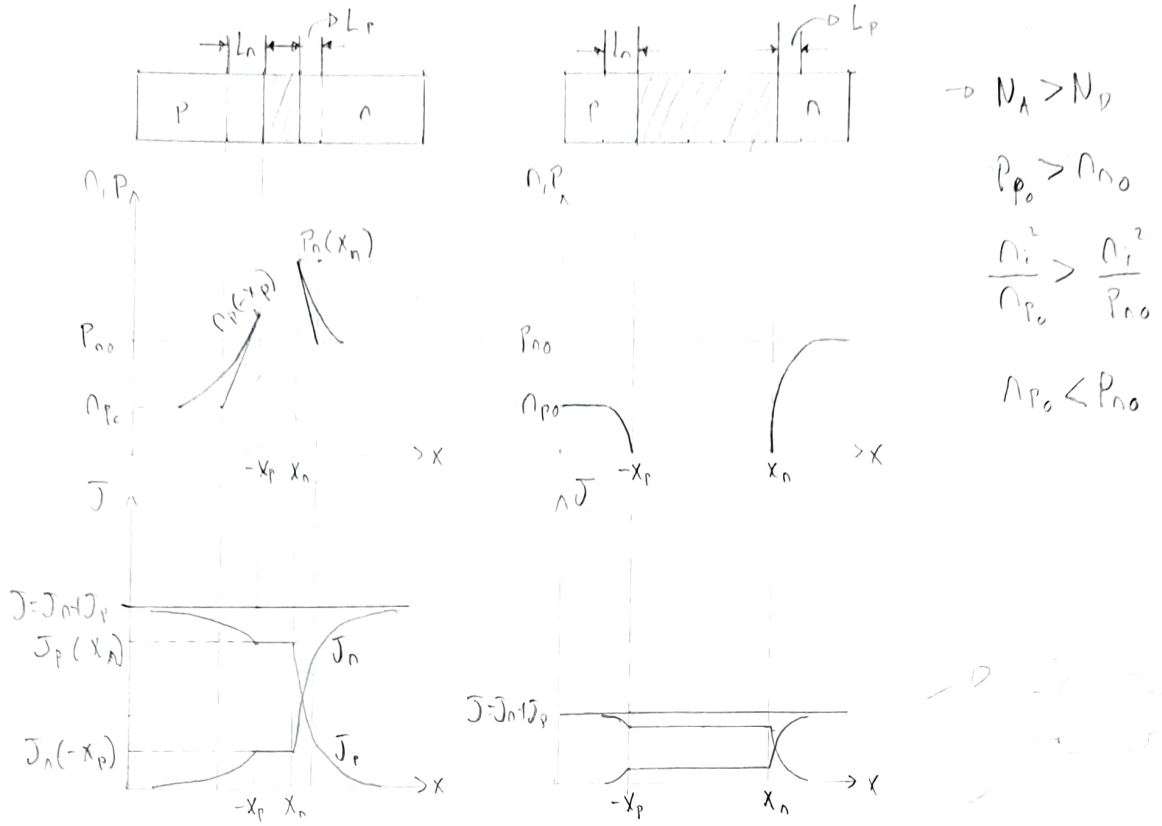
$$J_n(x = -x_p) = q D_n \left.\frac{dn_p}{dx}\right|_{x=-x_p}$$

positive in Fig.

$$J_n(x = -x_p) = \frac{q D_n n_{po}}{L_n}\left(e^{qV/kT} - 1\right) \left.e^{(x+x_p)/L_n}\right|_{x=-x_p}$$

$\uparrow$

The total current is constant throughout the device:



$\rightarrow N_A > N_D$

$p_{p0} > n_{n0}$

$\frac{n_i^2}{n_{p0}} > \frac{n_i^2}{p_{n0}}$

$n_{p0} < p_{n0}$

At thermal equilibrium:

$$J_n(x \leq -x_p) = \frac{q D_n n_{p0}}{L_n} \left(e^{qV/kT} - 1\right) e^{(x+x_p)/L_n} \rightarrow J_p(x \geq x_p) = \frac{q D_p p_{n0}}{L_p} \left(e^{qV/kT} - 1\right) e^{-(x-x_n)/L_p}$$

$J_p = 0$

$J_n = 0$

$$J = J_n + J_p = 0$$

Total current:

$$J = J_n(-x_p) + J_p(x_n) = \left(\frac{q D_n n_{p0}}{L_n} + \frac{q D_p p_{n0}}{L_p}\right)\left(e^{qV/kT} - 1\right)$$

Saturation current density:

$$\boxed{J_s = \frac{q D_n n_{p0}}{L_n} + \frac{q D_p p_{n0}}{L_p}}$$

Ideal diode equation:

$$\boxed{J = J_s\left(e^{qV/kT} - 1\right)}$$

In $x \leq -x_p$ ← From graph $J(x)$:

$$J_p(x \leq x_p) = J_s\left(e^{qV/kT} - 1\right) - J_n(x \leq -x_p)$$

$$J_p(x \leq -x_p) = J_s\left(e^{qV/kT} - 1\right) - \frac{q D_n n_{po}}{L_n}\left(e^{qV/kT} - 1\right) e^{(x+x_p)/L_n}$$

$$J_p(x \leq -x_p) = \left[J_s - \frac{q D_n n_{po}}{L_n} e^{(x+x_p)/L_n}\right]\left(e^{qV/kT} - 1\right)$$

In $x \geq x_n$: From graph $J(x)$:

$$J_n(x \geq x_n) = J_s\left(e^{qV/kT} - 1\right) - J_p(x \geq x_n)$$

$$J_p(x \geq x_n) = \left[J_s - \frac{q D_p p_{no}}{L_p} e^{-(x-x_n)/L_p}\right]\left(e^{qV/kT} - 1\right)$$

Ideal current-voltage characteristics:

$$\frac{J}{J_s} = e^{qV/kT} - 1$$

$$\text{Log}\left|\frac{J}{J_s}\right| = \text{Log}\left|e^{qV/kT} - 1\right|$$

For $V \geq 3kT/q$:

$$\text{Log}\left|\frac{J}{J_s}\right| = \text{Log}\left|e^{qV/kT} - 1\right| \approx \text{Log}\left|e^{qV/kT}\right| = (\text{Log}\, e)\frac{qV}{kT}$$

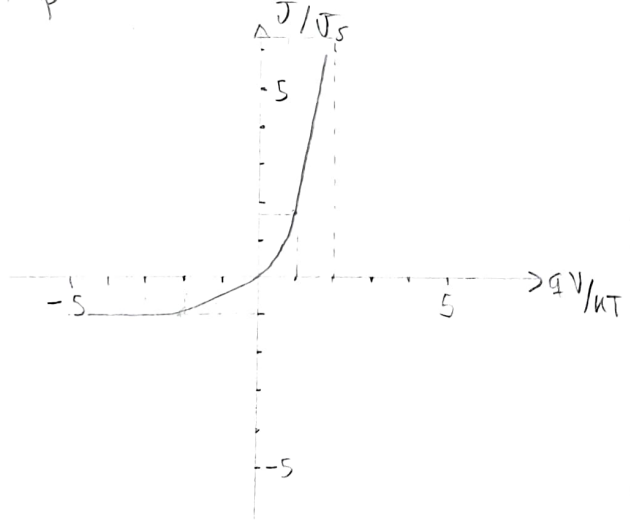
$e^{qV/kT} \gg 1$

$e^3 \gg 1$

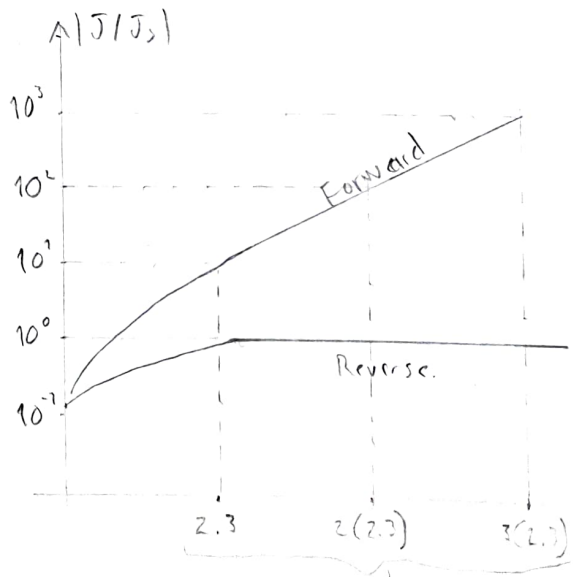
$20.1 \gg 1$

$$\text{Log}\left|\frac{J}{J_s}\right| = 0.434\frac{qV}{kT}$$

Cartesian plot:



Semilog plot:



see next page

For $V \geq 3kT/q$: ← Forward bias

$$\text{Log}\left(\frac{J}{J_s}\right) = 0.434 \frac{qV}{kT}$$

IF $\Delta \text{Log}\left(\frac{J}{J_s}\right) = 1$:

$$\Delta \text{Log}\left(\frac{J}{J_s}\right) = 0.434\, \Delta\left(\frac{qV}{kT}\right)$$

$\Delta(qV/kT) = 2.3$ → For every decade change of current

$$\frac{q\Delta V}{kT} = 2.3$$

$$\Delta V = 2.3\frac{kT}{q}$$

↓

The voltage change for an ideal diode is $\Delta V = \frac{2.3kT}{q}$

At $T = 300°K$:

$$\Delta V = 2.3\frac{(8.62 * 10^{-5}\ eV/°K)(300°K)}{e}$$

$\Delta V = 60mV$ → For every decade change of current

*For p^+-n junction:

$$J = \left(\frac{qD_n \cancel{n_{p0}}^{\to 0}}{L_n} + \frac{qD_p p_{no}}{L_p}\right)\left(e^{qV/kT} - 1\right)$$

$$J = \frac{qD_p N_V}{L_p}\left[e^{(qV - [E_F - E_V])/kT} - e^{-(E_F - E_V)/kT}\right]$$

The current is small if:

$$qV - (E_F - E_V) < 0$$

$$qV < E_F - E_V$$

$$V < \frac{E_F - E_V}{q} \longrightarrow \text{current increases rapidly}$$

if, slightly:

$$V > \frac{E_F - E_V}{q}$$

↓

cut-in voltage → Since, it is p_{no} → n-type:

↓

$E_F - E_V \cong E_g$

$$V > \frac{E_g}{q}$$

→ $p_0 = N_V e^{(E_V - E_F)/kT}$

$P_{0,} = N_V e^{-(E_F - E_V)/kT}$

↓

Effective density of states in the VB

$N_V = 2.66 * 10^{19} cm^{-3}$ - DS;

EXAMPLE 5

$A = 2 * 10^{-4}\ cm^2$ $N_A = 5 * 10^{16}\ cm^{-3}$ $N_D = 10^{16}\ cm^{-3}$ $n_i = 9.65 * 10^9\ cm^{-3}$

$D_n = 21\ cm^2/s$ $D_p = 10\ cm^2/s$

$\tau_p = \tau_n = 5 * 10^{-7}\ s$

I_s ideal = ? $\rightarrow P_{p0} = N_A$ $\rightarrow n_{n0} = N_D$

$$J_s = \frac{q D_p P_{n0}}{\sqrt{D_p \tau_p}} + \frac{q D_n n_{p0}}{\sqrt{D_n \tau_n}} = q\sqrt{\frac{D_p}{\tau_p}}\left(\frac{n_i^2}{N_D}\right) + q\sqrt{\frac{D_n}{\tau_n}}\left(\frac{n_i^2}{N_A}\right)$$

$$J_s = q n_i^2 \left(\frac{1}{N_D}\sqrt{\frac{D_p}{\tau_p}} + \frac{1}{N_A}\sqrt{\frac{D_n}{\tau_n}}\right)$$

$$J_s = (1.6 * 10^{-19}\ C)(9.65 * 10^9\ cm^{-3})^2 \left[\frac{1}{(10^{16}\ cm^{-3})}\sqrt{\frac{10\ cm^2/s}{5 * 10^{-7}\ s}} + \frac{1}{(5 * 10^{16}\ cm^{-3})}\sqrt{\frac{21\ cm^2/s}{5 * 10^{-7}\ s}}\right]$$

$$J_s = 8.59 * 10^{-12}\ C\ cm^{-6}\left(\frac{1}{cm^{-3}}\frac{cm}{s}\right)$$

$$J_s = 8.59 * 10^{-12}\ A\ cm^{-2}$$

$$\rightarrow I_s = J_s * A = (8.59 * 10^{-12}\ A/cm^2)(2 * 10^{-4}\ cm^2) = 1.72 * 10^{-15}\ A$$

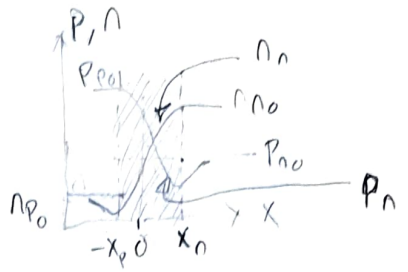
3.4.2 Generation-Recombination and High-Injection Effects

Ideal diode equation: $J = J_s(e^{qV/kT} - 1)$ → Germanium p-n junction at low current densities

↓ no in

Si, GaAs pn junctions ← Generation or recombination of carriers in the depletion region

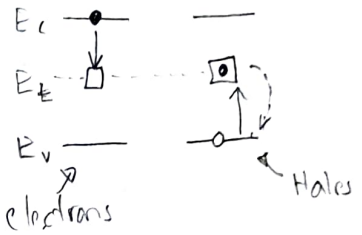
* Reverse-bias condition:



↳ Below their equilibrium

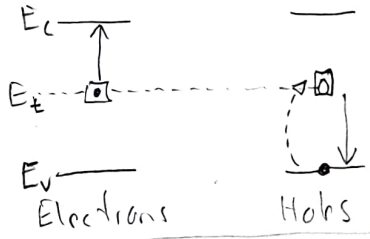
↓

The capture process aren't important: (Recombination)



↓

The emission process are important: (Recombination):



↳ In the steady states

Indirect recombination processes

↓

Si, GaAs

For indirect recombination:

$$U = \frac{V_{th}\,\sigma_n\,\sigma_p\,N_t\,(p_n n_n - n_i^2)}{\sigma_p\left[p_n + n_i e^{(E_i - E_t)/kT}\right] + \sigma_n\left[n_n + n_i e^{(E_t - E_i)/kT}\right]}$$

$[U] = cm^{-3}/s$

$[N_t] = cm^{-3}$

N_t: Concentration of the recombination center.

Electron-hole pair generation with: $p_n < n_i$ and $n_n < n_i$?:

$$G = -U = \frac{V_{th}\,\sigma_n\,\sigma_p\,N_t\,n_i}{\sigma_p e^{(E_i - E_t)/kT} + \sigma_n e^{(E_t - E_i)/kT}}$$

Generation life time:

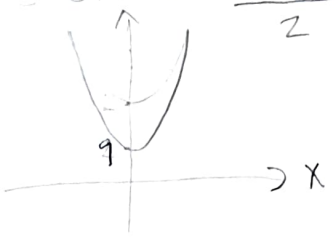
$$\tau_g = \frac{\sigma_p e^{(E_i - E_t)/kT} + \sigma_n e^{(E_t - E_i)/kT}}{\sigma_p \sigma_n v_{th} N_t}$$

$$\rightarrow G = -U = \frac{n_i}{\tau_g}$$

Simple case: $\sigma_n = \sigma_p = \sigma_0$:

$$G = \frac{v_{th} \sigma_0 N_t n_i}{2 \cosh\left(\frac{E_t - E_i}{kT}\right)}$$

$\cosh x = \frac{e^x + e^{-x}}{2}$



Maximum G at $E_t = E_i \rightarrow$ Middle of the band gap

$1/\cosh x = \frac{2}{e^x + e^{-x}}$



$$\downarrow$$

$$\tau_{g\,min} = \frac{2}{v_{th} \sigma_0 N_t} \cosh\left(\frac{E_t - E_i}{kT}\right) \rightarrow 1$$

$$\downarrow$$

Only carriers $E_t \approx E_i$: Contributes to the G $\quad [G] = \frac{cm^{-3}}{s}$

Current due to generation in the depletion region:

$$J_{gen} = \int_{-x_p}^{x_n} q G_{max} dx \approx q G_{max} W = \frac{q n_i W}{\tau_g} \rightarrow F(V_R)$$

For p^+-n $\rightarrow N_A \gg N_D$, and $V_R > 3kT/q \rightarrow V < -\frac{3kT}{q}$:

Diffusion current in the neutral regions from ideal diode equation:

$$J = \left(\frac{q D_p p_{n0}}{L_p} + \frac{q D_n n_{p0}}{L_n}\right)\left(e^{qV/kT} - 1\right)$$

$$J = \left(\frac{q D_p}{\sqrt{D_p \tau_p}} \frac{n_i^2}{N_D} + \frac{q D_n}{\sqrt{D_n \tau_n}} \frac{n_i^2}{N_A}\right)(-1)$$

$N_A \gg N_D$

$$J = -\left(q \sqrt{\frac{D_p}{\tau_p}} \frac{n_i^2}{N_D}\right)$$

Generation current in the depletion region:

$$J_{gen} = -\frac{q n_i W}{\tau_g}$$

- Current at reverse bias: $V < -\frac{3kT}{q}$:

$$J_R = -\left(q\sqrt{\frac{D_p}{\tau_p}}\frac{n_i^2}{N_D} + \frac{q n_i W}{\tau_g}\right)$$

$$|J_R| = q\sqrt{\frac{D_p}{\tau_p}}\frac{n_i^2}{N_D} + \frac{q n_i W}{\tau_g}$$

For semiconductors with large values of n_i eg. Ge:

$$J_R \approx -q\sqrt{\frac{D_p}{\tau_p}}\frac{n_i^2}{N_D}$$ → Diffusion current → ideal diode equation

If n_i is small e.g. Si and GaAs:

$$J_R \approx -\frac{q n_i W}{\tau_g}$$ → generation current in the depletion region

▷ EXAMPLE 6

By example 5: $N_A = 5*10^{16}\,cm^{-3}$ $D_n = 21\,cm^2/s$ Si: $n_i = 9.65*10^9\,cm^{-3}$

$N_D = 10^{16}\,cm^{-3}$ $D_p = 10\,cm^2/s$

$\tau_n = \tau_p = 5*10^{-7}\,s$

$J_s = 8.59*10^{-12}\,A\,cm^{-2}$ → J_{diff} } Ideal diode equation

Generation current density: ?

For $V = -4V = -V_R$

$V_R = 4V$

$\tau_g = \tau_n = \tau_p = 5*10^{-7}\,s$

$$J_{diff} = J_s\left(e^{qV/kT} - 1\right)$$

At $V = -V_R = -4V << 0{,}025V$

$$J_{diff} = -J_s = -8.59*10^{-12}\,A\,cm^{-2}$$

→ $W = \sqrt{\frac{2\varepsilon_s}{q}\left(\frac{N_A+N_D}{N_A N_D}\right)(V_{bi}-V)}$ ← Abrupt junction

$$W = \sqrt{\frac{2\varepsilon_s}{q}\left(\frac{N_A+N_D}{N_A N_D}\right)\left[\left(\frac{kT}{q}\ln\left(\frac{N_A N_D}{n_i^2}\right)\right)+V_R\right]}$$

$$W = \sqrt{\frac{2(11.9*553*10^{3}\not{e}/Vcm)}{\not{e}}\left[\frac{6*10^{16}cm^{-3}}{5*10^{32}cm^{-6}}\right]\left(25.9mV\ln\left(\frac{5*10^{32}cm^{-6}}{[9.65*10^{9}cm^{-3}]^2}\right)+V_R\right)}$$

$$W = 3.97*10^{-5}\frac{cm}{\sqrt{V}}\sqrt{0.759V+V_R}$$

→ $J_{gen}(V=-V_R=-4V) = -\frac{qn_iW}{\tau_g} = -\frac{(1.6*10^{-19}C)(9.65*10^{9}cm^{-3})W}{5*10^{-7}s}$

$$J_{gen} = -1.23*10^{-7}Acm^{-3}\left(\frac{cm}{\sqrt{V}}\right)\sqrt{0.759V+V_R}$$

$$J_{gen} = -1.23*10^{-7}A\,cm^{-2}\frac{}{\sqrt{V}}\sqrt{0.759V+4V}$$

$J_{gen} = -2.67*10^{-7}Acm^{-2}$

↑ Generation-recombination in scr.

→ $J_{diFF} = -8.59*10^{-12}A\,cm^{-2}$

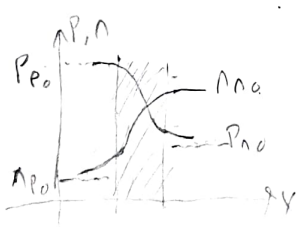
↑ Diffussion in neutral region.

$$|J_{gen}| >> |J_{diFF}|$$

$$\downarrow$$

$$J(-4V) = J_{gen}$$

* Under forward bias:



→ electrons and holes exceed their equilibrium values

↓

Carriers will attempt to return to equilibrium values by recombining.

↓

Capture process

At $x = x_n$:

$$p_n = p_{no} e^{qV/kT}$$

$$p_n n_n = p_{no} n_n e^{qV/kT}$$

$$p_n n_n = p_{no} n_{no} e^{qV/kT}$$

↑ Low injection

$$p_n n_n = n_i^2 e^{qV/kT}$$

From recombination for indirect:

$$U = \frac{v_{th} \sigma_n \sigma_p N_t (p_n n_n - n_i^2)}{\sigma_p [p_n + n_i e^{(E_i - E_t)/kT}] + \sigma_n [n_n + n_i e^{(E_t - E_i)/kT}]}$$

Assuming $\sigma_n = \sigma_p = \sigma_0$:

$$U = \frac{\sigma_0 v_{th} N_t n_i^2 (e^{qV/kT} - 1)}{n_n + p_n + 2 n_i \cosh\left(\frac{E_t - E_i}{kT}\right)} = -B$$



$U_{max} (E_t = E_i)$

↑ E_t near E_i

e.g. Au → $E_t - E_i = 0.02eV$ } In Si → $E_g = 1.12 eV$

Cu → $E_t - E_i = -0.02eV$ }

e.g. Cr → $E_t - E_i = 0.08 eV$ } In GaAs → $E_g = 1.42 eV$

For the case: $E_t = E_i$:

$$U = \sigma_0 v_{th} N_t \frac{n_i^2 (e^{qV/kT} - 1)}{n_n + p_n + 2 n_i}$$

$n_n + p_n$ minimum → U_{max}

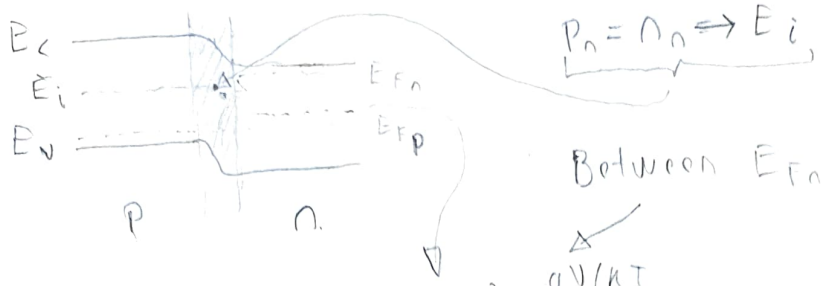
Localization in the depletion region

$$d(p_n + n_n) = 0$$

$$dp_n = -dn_n$$

$$dp_n = -d\left(\frac{\overbrace{p_n n_n}^{n_i^2 e^{qV/kT} * F(\text{localization})}}{p_n}\right) = \frac{1}{p_n^2} p_n n_n dp_n$$

$p_n = n_n$



$p_n = n_n \Rightarrow E_i$

Between E_{Fn} and E_{Fp}

$$p_n n_n = n_i^2 e^{qV/kT}$$

If $p_n = n_n$:

$$p_n = n_n = \sqrt{n_i^2 e^{qV/kT}}$$

$$p_n = n_n = n_i e^{qV/2kT} \leftarrow \text{Where } U_{max}.$$

$$U_{max} = \frac{\sigma_0 v_{th} N_t n_i^2 (e^{qV/kT} - 1)}{p_n + n_n + 2n_i}$$

$$U_{max} = \frac{\sigma_0 v_{th} N_t n_i^2 (e^{qV/kT} - 1)}{2n_i (e^{qV/2kT} + 1)}$$

For $V > 3kT/q \rightarrow V > 3(25.9\,mV) \rightarrow V > 77.7\,mV$:

$\rightarrow e^3 = 20 > 1$

$\rightarrow e^{3/2} = 4.48 > 1$

$$U_{max} = \frac{\sigma_0 v_{th} N_t n_i (e^{qV/kT} \cancel{-1})}{2 (e^{qV/2kT} \cancel{+1})}$$

$$U_{max} = \frac{1}{2} \sigma_0 v_{th} N_t n_i e^{qV/2kT}$$

Recombination current:

$$J_{rec} = \int_{-x_p}^{x_n} q U_{max} dx \approx \frac{qW}{2} \sigma_0 v_{th} N_t n_i e^{qV/2kT}$$

Effective recombination lifetime:

$$\tau_r = \frac{1}{\sigma_0 v_{th} N_t}$$

$$\rightarrow J_{rec} = \frac{qW n_i}{2\tau_r} e^{qV/2kT}$$

Total forward current. For $p_{no} \gg n_{po}$ and $V > 3kT/q$

$$\frac{n_i^2}{N_D} \gg \frac{n_i^2}{N_A} \qquad V > 77.7\,mV$$

$N_A \gg N_D \leftarrow p^+ - n$

$p^+ - n$

$\rightarrow J_{diff} = J_s\left(e^{qV/kT} - 1\right)$ $\rightarrow e^3 = 20.1$

$$J_{diff} = q\sqrt{\frac{D_p}{\tau_p}}\frac{n_i^2}{N_D}e^{qV/kT}$$

$\rightarrow f\left(\sqrt{V_E}, V_F\right)$ $\rightarrow$ W reduce at high V_F $\rightarrow J_{diff} > J_{rec}$ $\rightarrow n \simeq 1$

$$J_F = J_{diff} + J_{rec} = q\sqrt{\frac{D_p}{\tau_p}}\frac{n_i^2}{N_D}e^{qV/kT} + \frac{qWn_i}{2\tau_r}e^{qV/2kT}$$

Empirically:

$$J_F \approx \exp\left(\frac{qV}{nkT}\right)$$

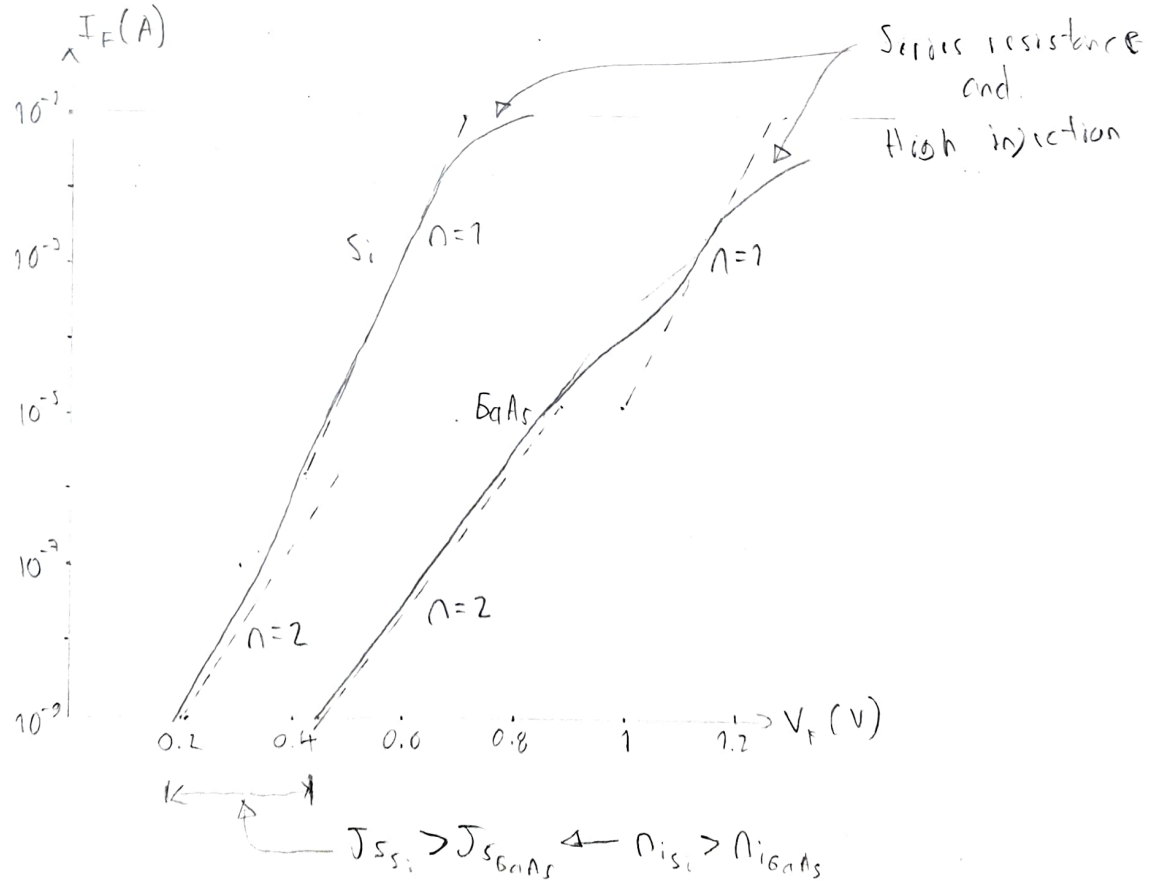
$\rightarrow n$: idrality factor

$n = 1 \leftarrow J_{diff} > J_{rec}$

$n = 2 \leftarrow J_{rec} > J_{diff}$

J_{rec} near $J_{diff} \rightarrow 1 < n < 2$

Forward current-voltage at 300K:



At low and medium current levels:

IR drop across the neutral regions $< \frac{kT}{q} \rightarrow 26\,mV$ at 300 K

e.g.: S_1: $R = 1.5\,\Omega$ at 1 mA

$IR = 1.5\,mV < 26\,mV$

However at 100 mA

$IR = 150\,mV = 0.15\,V > 26\,mV$

$0.15\,V = 5.7\,\frac{kT}{q}$

This IR drop reduces the bias across the depletion region:

$$I \approx I_s \exp\left[\frac{q(V - IR)}{nkT}\right] = \frac{I_s \exp(qV/nkT)}{\exp\left(\frac{qIR}{nkT}\right)} = \frac{I_{diff}}{\exp\left(q\frac{IR}{nkT}\right)}$$

(I_{diff}: ideal diff. current)

$n = 1$ if $V_F >> \frac{kT}{q}$

At high current densities: Injected minority carrier density is comparable to the majority carrier concentration at the n-side:

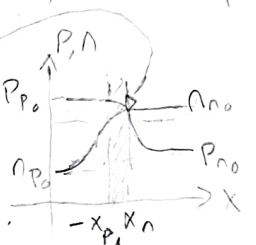
$P_n(x = x_n) \approx n_n$

At $x = x_n$: $P_n = n_n$

$P_n n_n = \left(P_{no} e^{qV/kT}\right) n_{no}$

$P_n n_n = n_i^2 e^{qV/kT}$

$n_n = P_n = n_i e^{qV/2kT}$ At $x = x_n$



No into the scr. like low injection for calculation of U_{max}

Using this as a boundary condition: $J_{diff} \propto \exp(qV/2kT)$

↓

J_{diff} increase at a slower rate

↓

Continuity equation?

3.4.3 Temperature Effect

* Forward-bias case:

Hole diffusion current to the recombination

$$\frac{J_{diff}}{J_{rec}} = \frac{\left[\frac{q D_p P_{no}}{L_p}\left(e^{qV/\eta kT} - 1\right)\right]}{\frac{q W n_i}{2\tau_r} e^{qV/2kT}} = \frac{2\tau_r D_p n_i^2}{L_p W N_D} e^{qV/2kT}$$

n_{no}

$$\frac{J_{diff}}{J_{rec}} = \frac{2\tau_r n_i D_p}{L_p N_D W} e^{qV/2kT}$$

$$\rightarrow L_p = \sqrt{D_p \tau_p}$$

$$\frac{L_p^2}{\tau_p} = D_p$$

$$\frac{J_{diff}}{J_{rec}} = 2\frac{n_i}{N_D}\frac{L_p}{W}\frac{\tau_r}{\tau_p} e^{qV/2kT}$$

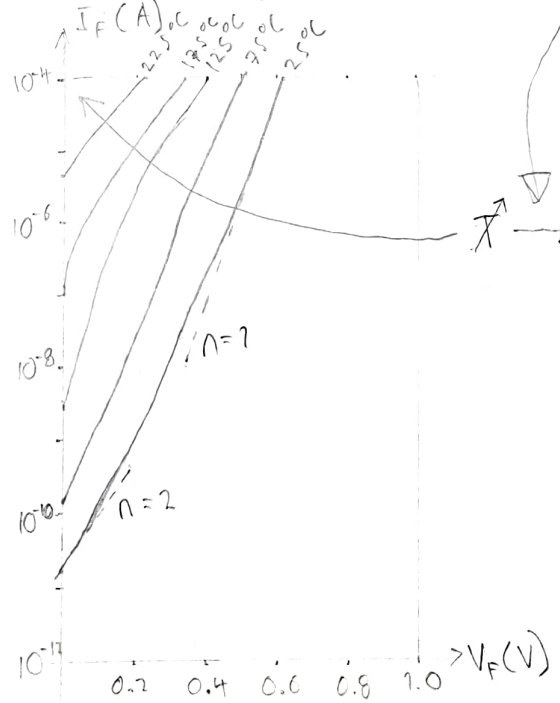
$$\rightarrow n_i = \sqrt{N_c N_v}\, \exp\left(-\frac{E_g}{2kT}\right)$$

Effective density of states in the conduction and valence bands

$$\frac{J_{diff}}{J_{rec}} \propto \exp\left[-\frac{(E_g - qV)}{2kT}\right]$$

While: $qV < E_g$

Forward characteristics of a Si diode:



$$\rightarrow \frac{I_{diff}}{I_{rec}}$$

For p^+-n in which: $I_{diff} >> I_{rec}$:

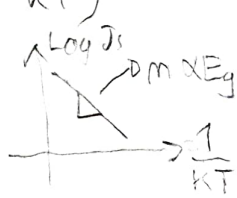
$$J_s = \frac{q D_p n_i^2}{L_p N_D} \propto \left[\exp\left(-\frac{E_g}{2kT}\right)\right]^2$$

n_{no}

Activation energy

$$J_s \propto \exp\left(-\frac{E_g}{kT}\right)$$

$$\text{Log } J_s \propto \frac{1}{kT}$$

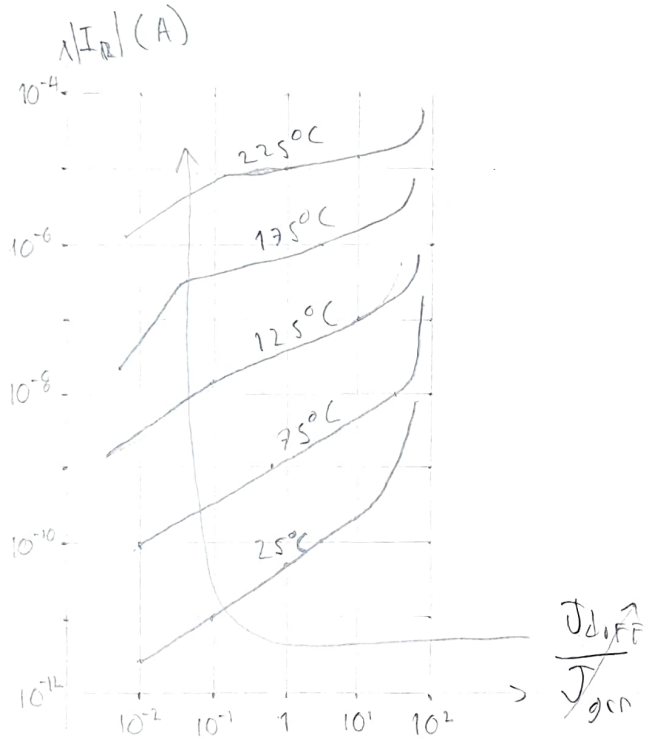


* Reverse-bias condition: p^+-n: $V \ll 0$

$$\frac{J_{diFF}}{J_{gen}} = \frac{\frac{q}{L_p}\left(\frac{L_p}{\tau_p}\right)\frac{n_i^2}{N_D}\left(e^{qV/kT} - 1\right)}{-\frac{q\,n_i\,W}{\tau_g}}$$

$$\frac{J_{diFF}}{J_{gen}} = \frac{n_i}{N_D}\frac{L_p}{W}\frac{\tau_g}{\tau_p} = \frac{1}{N_D}\frac{L_p}{W}\frac{\tau_g}{\tau_p}\sqrt{N_C N_V}\exp\left(-\frac{E_g}{2kT}\right)$$

Reverse bias: Si diode



At low T.: $J_{gen} > J_{diFF}$

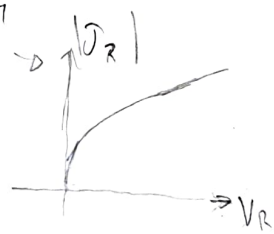
$$J_R = J_{gen}$$

Reverse.

$$|J_R| \approx \frac{q\,n_i\,W}{\tau_g} \qquad W \propto \sqrt{V_{bi} + V_R}$$

$$W \propto \sqrt{V_R}$$

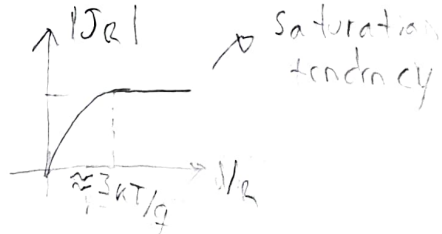
$$|J_R| \propto \sqrt{V_R}$$



At $T > 175°C$, $V_R \gtrsim 3kT/q$: $J_{diFF} > J_{gen}$

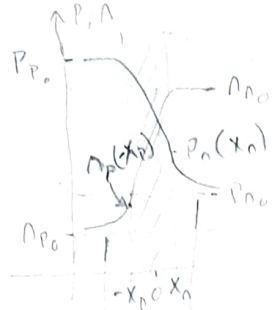
$$J_R = J_{diFF},\ V_R \gtrsim 3kT/q$$

$$|J_R| = cte$$



3.5 CHARGE STORAGE AND TRANSIENT BEHAVIOR

Under forward bias:



→ Minority carriers, injected across the junction, recombine with the majority carriers and decay exponentially

→ current flow (diffusion)

→ charge storage

3.5.1 Minority-Carrier Storage

Injected minoritary carriers per unit area stored in the neutral n-region:

$$Q_p = q\int_{x_n}^{\infty}(p_n - p_{n0})dx \rightarrow [Q_p] = C\,cm^{-2}$$

$$Q_p = q\int_{x_n}^{\infty}\left[p_{n0}\left(e^{qV/kT}-1\right)e^{-(x-x_n)/L_p}\right]dx$$

$$\rightarrow \int_{x_n}^{\infty} e^{(-x+x_n)/L_p}\,dx = -L_p e^{(-x+x_n)/L_p}\Big|_{x=x_n}^{x\to\infty}$$

$$= L_p$$

$$\rightarrow Q_p = q L_p p_{n0}\left(e^{qV/kT}-1\right)$$

$$Q_p = q L_p\left(p_{n0}e^{qV/kT} - p_{n0}\right)$$

$$Q_p = q L_p\left[p_n(x_n) - p_{n0}\right]$$

In terms of the injected current:

$$Q_p = \left(\frac{L_p^2}{D_p}\right)\frac{q D_p p_{n0}}{L_p}\left(e^{qV/kT}-1\right)$$

$$Q_p = \tau_p J_{p_{diff}}(x_n)$$ ← Average lifetime of holes in n-side.

Q_p is replenished every τ_p seconds

Injected minoritary carriers per unit area stored in the neutral p-region

$$Q_n = q\int_{-\infty}^{-x_p}(n_p - n_{p_0})\,dx \rightarrow [Q_n] = C\,cm^{-2}$$

$$Q_n = q\int_{-\infty}^{-x_p}\left[n_{p_0}\left(e^{qV/kT}-1\right)e^{(x+x_p)/L_n}\right]dx$$

$$\rightarrow \int_{-\infty}^{-x_p} e^{(x+x_p)/L_n}\,dx = L_n e^{(x+x_p)/L_n}\Big|_{-\infty}^{-x_p}$$

$$= L_n$$

$$\rightarrow Q_n = q L_n n_{p_0}\left(e^{qV/kT}-1\right)$$

$$Q_n = q L_n\left(n_{p_0}e^{qV/kT} - n_{p_0}\right)$$

$$Q_n = q L_n\left[n_p(-x_p) - n_{p_0}\right]$$

In terms of the injected current:

$$Q_n = \left(\frac{L_n^2}{D_n}\right)\frac{q p_n n_{p_0}}{L_n}\left(e^{qV/kT}-1\right)$$

$$Q_n = \tau_n J_{n_{diff}}(-x_p)$$ ← Average lifetime of electrons in p-side

Q_n is replenished every τ_n seconds

▷ EXAMPLE 7

Si, p^+-n $N_D = 8 * 10^{15}\ cm^{-3}$ Q_p = ? in the neutral n-region

$V_F = 1V$ $L_p = 5\ \mu m$

$$\rightarrow Q_p = qL_p p_{no}\left(e^{qV/kT} - 1\right) = qL_p \frac{n_i^2}{N_D}\left(e^{qV/kT} - 1\right)$$

$$Q_p = e\,(5 * 10^{-4}\ cm)\,\frac{(9.65 * 10^{9}\ cm^{-3})^2}{(8 * 10^{15}\ cm^{-3})}\left[e^{1eV/0.0259eV} - 1\right]$$

$$Q_p = 341 * 10^{15}\ e\ cm^{-2} = 54.6 * 10^{-3}\ \frac{C}{cm^2} = 5.46 * 10^{-2}\ C/cm^2$$

3.5.2 Diffusion Capacitance

Q_p, Q_n → Additional capacitance at forward biased

Diffusion capacitance of the stored holes in the neutral n-region: $[C_d] = F$

$$C_d = A\frac{dQ_p}{dV} = A\frac{d}{dV}\left(qL_p p_{no}\left[e^{qV/kT} - 1\right]\right)$$

$$\boxed{C_d = \frac{Aq^2 L_p p_{no}}{kT} e^{qV/kT}}$$

Diffusion capacitance of the stored electrons in neutral p-region: $[C_d] = F$

$$C_d = \frac{A\,dQ_n}{dV} = A\frac{d}{dV}\left(qL_n n_{p_o}\left[e^{qV/kT} - 1\right]\right)$$

$$\boxed{C_d = \frac{Aq^2 L_n n_{p_o} e^{qV/kT}}{kT}}$$

For p^+-n $N_A \gg N_D$

$$\frac{1}{N_A} \ll \frac{1}{N_D}$$

$$\frac{n_i^2}{N_A} \ll \frac{n_i^2}{N_D}$$

$n_{p_o} \ll p_{no} \rightarrow C_d \approx 0$ of the stored electrons

Under reverse bias:

$V < 0 \rightarrow e^{qV/kT} \approx 0 \rightarrow C_d \approx 0$

Conductance of ideal diode: When $J_{diff} \gg J_s$:

$$G = A\frac{dJ_{diff}}{dV} = A\frac{d}{dV}\left[J_s\left(e^{qV/kT} - 1\right)\right] = \frac{qA}{kT} J_s\left(e^{qV/kT}\right)$$

$$G = \frac{qA}{kT}\left[J_s\left(e^{qV/kT} - 1 + 1\right)\right] = \frac{qA}{kT}\left(J_{diff} + J_s\right) \cong \frac{qI_{diff}}{kT}$$

$$G \cong \frac{q I_{diff}}{nT} \qquad [G] = 1/\Omega$$

Depletion capacitance:

$$C_j = \frac{\varepsilon_s A}{W} \qquad [C_j] = F$$

For low-voltage, sinusoidal excitation

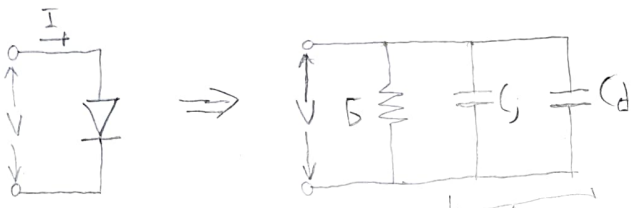
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Diode small-signal

⇓

By C_j and C_d → Diode: Biased quiescently → i.e. at DC

Diode small-signal equivalent circuit:



Q_j in the middle of Q_n and Q_p → Q_d

⇓

C_j parallel to C_d